ESE 4011 Complete Study Pack

Materials for Energy Engineering Application · KUET · past papers 2022 · answers from standard textbooks (no course slides available)

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 4 h for Part A and 4 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2022 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Only one paper (2022) exists, so every topic has the same chance; the plan therefore puts easy topics first. The fuel cell block (A7–A12) is one big family: learn the comparison table in the A10 memory box and you can answer most of Section A Q3 and Q4. The two numericals (B2 capacitors, B9 XRD) are quick sure marks with the calculators.

Section A: Solar cell materials and fuel cells

Block 1: very likely and easy · 3.8 h

Topic	Chance	Level	Covered
A8 Fuel cell stack (bipolar plates, series connection); DMFC layer structure	Very likely 100%	Easy	9%
A11 Phosphoric acid fuel cell (PAFC): electrolyte properties and functions, electrodes and catalyst	Very likely 100%	Easy	19%
A2 Crystalline silicon structures (mono-, multi-, amorphous); diffusion and drift device concepts	Very likely 100%	Easy	27%
A5 Chalcopyrite materials (CIS, CIGS) and the CdTe cell: layer structure and working principle	Very likely 100%	Easy	36%
A9 Role of PTFE in alkaline fuel cells; functions of bipolar plates in PEMFCs	Very likely 100%	Easy	44%
A12 Solid oxide fuel cell (SOFC): favourable characteristics, electrolyte types	Very likely 100%	Easy	53%
A4 A third-generation thin-film cell: dye-sensitised solar cell (working principle)	Very likely 100%	Easy	60%
A7 Fuel cell: definition, difference from a battery, working principle	Very likely 100%	Easy	67%
A10 Molten carbonate fuel cell (MCFC): construction, principle, features	Very likely 100%	Easy	74%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered
A3 High-efficiency concepts of c-Si cells; the PERL cell	Very likely 100%	Medium	84%
✔ Pass line: 84% of Section A marks covered, enough to choose 3 of the 4 questions (about 4.5 h)			
A6 Multijunction cells: challenges; spectral conversion	Very likely 100%	Medium	93%
A1 Losses in solar cells; optical losses	Very likely 100%	Medium	100%

Section B: Supercapacitors, hydrogen storage, thin films, XRD and electron microscopy

Block 1: very likely and easy · 2.9 h

Topic	Chance	Level	Covered
B10 SEM: backscattered vs secondary electrons; how SEM gives high resolution and depth of field	Very likely 100%	Easy	13%
B6 Chemical vapour deposition (CVD): process and role in thin films, powders, single crystals; PVD vs CVD	Very likely 100%	Easy	24%
B5 Sputtering: definition and process	Very likely 100%	Easy	31%
B8 Geometry and components of an X-ray diffractometer	Very likely 100%	Easy	37%
B7 Obtaining monochromatic Kα radiation from the X-ray tube spectrum	Very likely 100%	Easy	43%
B3 Pseudocapacitor vs double-layer capacitor	Very likely 100%	Easy	48%
B2 Numerical: capacitors 15 F and 8 F in series and in parallel across 50 V	Very likely 100%	Easy	52%

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
B4 Hydrogen storage in carbonaceous materials and CNTs: adsorption mechanism, key parameters, applications	Very likely 100%	Medium	66%
B9 XRD numericals: 2θ of Co (111) with Cu Kα; structure factor of an FCC cell	Very likely 100%	Medium	79%
✔ Pass line: 79% of Section B marks covered, enough to choose 3 of the 4 questions (about 4.4 h)			
B11 TEM: diffraction pattern and image together; preparing a 0.1 mm ceramic sample	Very likely 100%	Medium	91%

B1 Electric double-layer capacitor (EDLC): why high capacitance, fundamental principle	Very likely 100%	Medium	100%

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3. **Section B: Supercapacitors, hydrogen storage, thin films, XRD and electron microscopy**: B10 SEM: backscattered vs secondary electrons; how SEM gives high resolution and depth of field, B6 Chemical vapour deposition (CVD): process and role in thin films, powders, single crystals; PVD vs CVD, B5 Sputtering: definition and process, B8 Geometry and components of an X-ray diffractometer, B7 Obtaining monochromatic Kα radiation from the X-ray tube spectrum, B3 Pseudocapacitor vs double-layer capacitor, B2 Numerical: capacitors 15 F and 8 F in series and in parallel across 50 V, B4 Hydrogen storage in carbonaceous materials and CNTs: adsorption mechanism, key parameters, applications, B9 XRD numericals: 2θ of Co (111) with Cu Kα; structure factor of an FCC cell, B11 TEM: diffraction pattern and image together; preparing a 0.1 mm ceramic sample, B1 Electric double-layer capacitor (EDLC): why high capacitance, fundamental principle

One-shot notes: ESE 4011 Materials for Energy Engineering Application

Solar cell losses

Optical: reflection (bare Si 30-40 %, fix with ARC λ/4n + texture), shading by fingers (5-10 %), transmission (light trapping, back reflector).

Spectral: non-absorption ($E < E_g$) and thermalisation ($E - E_g$ as heat). **Recombination**: radiative, Auger, SRH, surface (passivate). **Resistive**: series and shunt (fill factor).

c-Si and devices

Mono (one crystal, Cz/FZ, 20-26 %), multi (grains, cast, 16-22 %), amorphous a-Si:H (no order, PECVD, E_g 1.7 eV, 6-10 %).

Diffusion device = p-n, pure material, carriers diffuse (c-Si). **Drift device** = p-i-n, field in i-layer sweeps carriers (a-Si).

PERL (UNSW, 25 %)

Inverted pyramids, double ARC on thin oxide, light n^+ emitter + heavy n^{++} under fingers, rear oxide passivation + reflector, local p^+ point contacts, FZ wafer. Led to PERC, PERT, TOPCon, IBC.

Thin films

DSSC: FTO / dye on mesoporous TiO_2 / I^-/I_3^- electrolyte / Pt counter. Dye excites → injects e^- into TiO_2 → load → Pt reduces I_3^- → I^- regenerates dye.

Chalcopyrite I-III-VI_2: $CuInSe_2$ 1.0, $CuGaSe_2$ 1.7, $CuInS_2$ 1.5 eV, CIGS 1.1-1.2 eV.

CdTe (superstrate): glass / TCO / buffer / n-CdS (2.4 eV) / p-CdTe (1.45 eV, $CdCl_2$ treated) / back contact.

Beyond one junction

Multijunction: GaInP/GaAs/Ge, gaps decrease downward; > 47 % with concentration. Challenges: current matching, spectrum change, lattice matching, tunnel junctions, cost.

Spectral conversion: down-conversion (1 → 2 photons, cuts thermalisation), down-shifting (UV → visible), up-conversion (2 IR → 1, uses sub-gap light).

Fuel cell basics

Converts fuel chemical energy to electricity continuously; battery stores. Anode: $H_2 \rightarrow 2H^+ + 2e^-$; cathode: $½O_2 + 2H^+ + 2e^- \rightarrow H_2O$. E^0 = 1.23 V, real 0.6-0.8 V.

Stack: series via bipolar plates, gases in parallel via manifolds, end plates + tie rods. **Bipolar plate**: connect, distribute, separate, drain water, cool, support.

PTFE in AFC: hydrophobic, stops flooding, keeps three-phase boundary, binder.

Fuel cell types table

Type	Electrolyte	Ion	T (°C)	Catalyst
PEMFC	Nafion	H^+	60-80	Pt
DMFC	Nafion	H^+	60-120	Pt-Ru / Pt
AFC	KOH	OH^-	60-200	Ni, Ag
PAFC	H_3PO_4 in SiC	H^+	150-220	Pt/C
MCFC	Li/K carbonate in $LiAlO_2$	CO_3^{2-}	650	Ni
SOFC	YSZ ceramic	O^{2-}	600-1000	Ni-YSZ / LSM

DMFC, MCFC, PAFC, SOFC

DMFC layers: plate / GDL / Pt-Ru / Nafion / Pt / GDL / plate. $CH_3OH + H_2O \rightarrow CO_2 + 6H^+ + 6e^-$. Problem: methanol crossover.

MCFC: anode $H_2 + CO_3^{2-} \rightarrow H_2O + CO_2 + 2e^-$; cathode $½O_2 + CO_2 + 2e^- \rightarrow CO_3^{2-}$; CO_2 recycled.

PAFC acid: proton conductor, stable, low volatility, CO_2 tolerant, freezes at 42 °C. Electrodes: Pt/C + PTFE on carbon paper.

SOFC: efficient, fuel flexible, no Pt, all solid, useful heat. Electrolytes: YSZ, ScSZ, GDC/SDC, LSGM, proton perovskites.

Supercapacitors

EDLC: porous carbon + electrolyte; ions form a double layer (d < 1 nm), huge A → C = εA/d large. Electrostatic, 10^6 cycles. E = ½CV^2.

Pseudocapacitor: fast surface redox (RuO_2, MnO_2, PANI, PPy); adsorption, redox, intercalation.

Numerical: series C_{eq} = 120/23 = 5.217 F, Q = 260.87 C, V_{15F} = 17.39 V, V_{8F} = 32.61 V; parallel 23 F, 50 V each.

H_2 in carbon

Physisorption (weak, 77 K, ~1 wt % per 500 m^2/g), chemisorption (C-H, needs heat), spillover (Pt/Pd). Increase: area, pores 0.6-0.7 nm, B/N doping, metal decoration, cool. CNT sites: outside, inside, interstitial, grooves; < 1 wt % at room T.

Thin films

Sputtering: vacuum, Ar, plasma, Ar^+ hits target, atoms ejected, deposit. DC (conductors), RF (insulators), magnetron (fast), reactive.

CVD steps: transport, diffuse, adsorb, react, desorb. SiH_4 → Si + $2H_2$. PVD = physical, line-of-sight, low T; CVD = chemical, conformal, higher T, toxic gases.

XRD

$n\lambda = 2d \sin\theta$; $d = a/\sqrt{(h^2+k^2+l^2)}$. Cu Kα 1.5418 Å, Kβ 1.392 Å; **Ni filter** (edge 1.488 Å) or crystal monochromator.

Co (111): d = 2.0461 Å, θ = 22.13°, 2θ = 44.27°.

FCC: $F = f[1 + e^{\pi i(h+k)} + e^{\pi i(h+l)} + e^{\pi i(k+l)}]$: unmixed → 4f, mixed → 0. Present 111, 200, 220, 311, 222. BCC: h+k+l even.

Diffractometer: tube, slits, sample at θ, detector at 2θ, focusing circle.

Electron microscopy

SE: < 50 eV, 1-10 nm deep, topography, Everhart-Thornley. **BSE**: high energy, elastic, Z contrast. SEM: gun, condenser, scan coils, objective, sample, detectors.

TEM: intermediate lens on image plane = image; on back focal plane = diffraction (SAED). Bright field / dark field via objective aperture.

Ceramic prep: saw, 3 mm disc, grind 100 µm, dimple 10-20 µm, Ar ion mill, carbon coat (or FIB).

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
µc-Si:H	hydrogenated microcrystalline silicon
a-Si:H	hydrogenated amorphous silicon
AFC	alkaline fuel cell
ALD	atomic layer deposition
APCVD	atmospheric-pressure CVD
ARC	anti-reflection coating
BSE	backscattered electrons
c-Si	crystalline silicon
CGS	copper gallium selenide
CHP	combined heat and power
CIGS	copper indium gallium (di)selenide, $Cu(In,Ga)Se_2$
CIS	copper indium (di)selenide, $CuInSe_2$
CNT	carbon nanotube
CO	carbon monoxide
CV	cyclic voltammetry
CVD	chemical vapour deposition
Cz	Czochralski (crystal pulling)
CZTS	copper zinc tin sulphide (kesterite)
DC	direct current
DMFC	direct methanol fuel cell
DOE	US Department of Energy
DSSC	dye-sensitised solar cell
EBSD	electron backscatter diffraction
EDLC	electric double-layer capacitor
EDS	energy-dispersive X-ray spectroscopy
EDX	energy-dispersive X-ray (analysis)
Eg	band gap energy
ESR	equivalent series resistance
eV	electron-volt (energy unit, 1.6×10^{-19} J)
FEG	field-emission gun
FIB	focused ion beam
FTO	fluorine-doped tin oxide
FZ	float zone (refining)
GDC	gadolinium-doped ceria
GDL	gas diffusion layer

HIT	heterojunction with intrinsic thin layer
HRTEM	high-resolution TEM
IBC	interdigitated back contact
ICDD	International Centre for Diffraction Data
IR	infrared
ITO	indium tin oxide
IUPAC	International Union of Pure and Applied Chemistry
JCPDS	Joint Committee on Powder Diffraction Standards
LPCVD	low-pressure CVD
LSCF	lanthanum strontium cobalt ferrite
LSGM	strontium- and magnesium-doped lanthanum gallate
LSM	strontium-doped lanthanum manganite
mc-Si	multicrystalline silicon
MCFC	molten carbonate fuel cell
MEA	membrane electrode assembly
MOCVD	metal-organic CVD
MOF	metal-organic framework
MWCNT	multi-walled carbon nanotube
PAFC	phosphoric acid fuel cell
PANI	polyaniline
PCFC	proton-conducting ceramic fuel cell
PECVD	plasma-enhanced chemical vapour deposition
PEDOT	poly(3,4-ethylenedioxythiophene)
PEM	proton exchange (polymer electrolyte) membrane
PEMFC	proton exchange membrane fuel cell
PERC	passivated emitter and rear cell
PERL	passivated emitter, rear locally-diffused (cell)
PERT	passivated emitter, rear totally-diffused
PLD	pulsed laser deposition
PPy	polypyrrole
PTFE	polytetrafluoroethylene (Teflon)
PV	photovoltaic
PVD	physical vapour deposition
RF	radio frequency
SAED	selected-area electron diffraction
ScSZ	scandia-stabilised zirconia
SDC	samarium-doped ceria

SE	secondary electrons
SEM	scanning electron microscope
SHJ	silicon heterojunction
SOEC	solid oxide electrolysis cell
SOFC	solid oxide fuel cell
SWCNT	single-walled carbon nanotube
TCO	transparent conducting oxide
TEM	transmission electron microscope
TOPCon	tunnel oxide passivated contact
UNSW	University of New South Wales
UV	ultraviolet
wt %	percent by weight
XRD	X-ray diffraction
YSZ	yttria-stabilised zirconia

Section A: Solar cell materials and fuel cells

A8 Fuel cell stack (bipolar plates, series connection); DMFC layer structure

Must study · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2022 Q3(b)** [13] How are the individual fuel cells connected within a fuel cell stack? Provide a brief explanation of the layer structure of a Direct Methanol Fuel Cell (DMFC).

★ REMEMBER IT

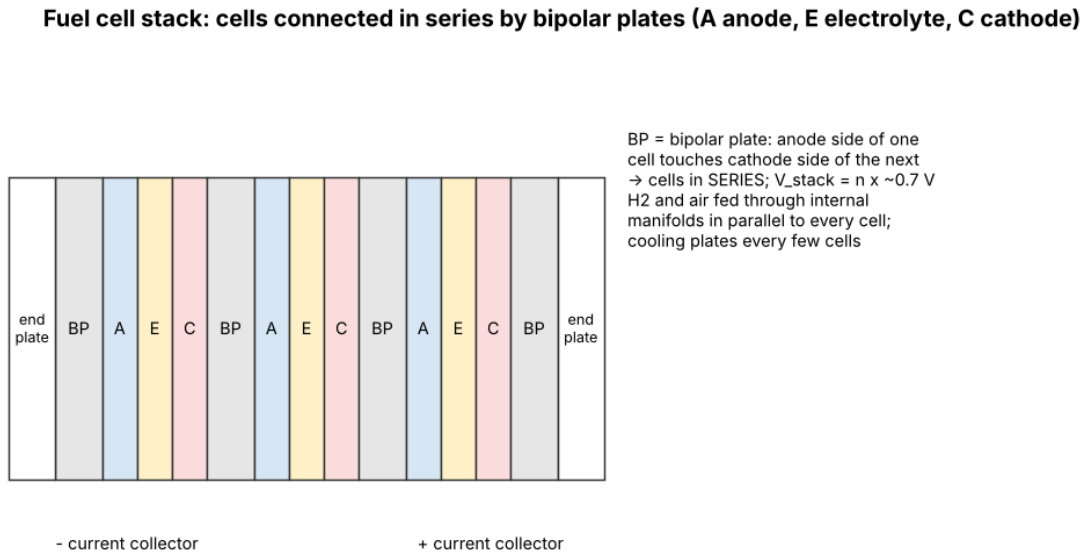
- Stack = cells in **series** through **bipolar plates**: one face is the anode of a cell, the other face the cathode of the next, so voltages add (about 0.7 V per cell).
- DMFC (direct methanol fuel cell) layers: **bipolar plate → anode diffusion layer → anode catalyst (Pt-Ru) → Nafion membrane → cathode catalyst (Pt) → cathode diffusion layer → bipolar plate**. Methanol + water at anode gives CO_2, protons and electrons.

Before you start: why stack cells

One fuel cell gives only about 0.7 V under load - too little for any practical use. Like batteries in a torch, many cells are connected in **series** to add their voltages; the result is a **stack**. The trick is the **bipolar plate**: one plate touches the anode of one cell on one side and the cathode of the next cell on the other side.

A **DMFC** (direct methanol fuel cell) feeds liquid methanol directly instead of hydrogen gas - easy to store, good for portable devices.

How individual cells are connected in a stack



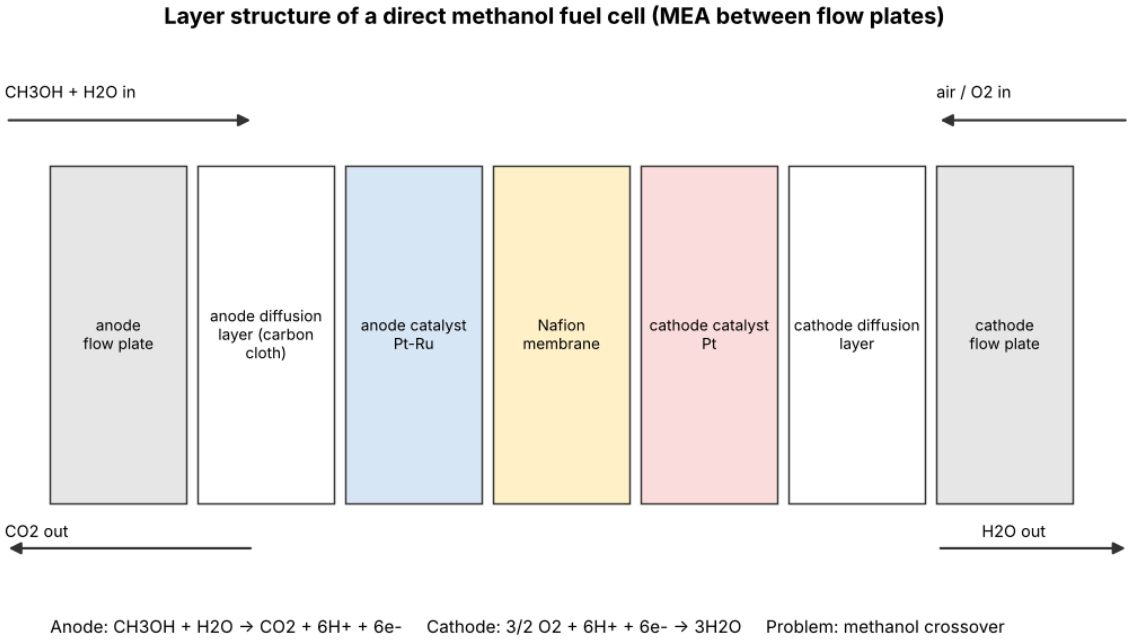
Fuel cell stack with bipolar plates (drawn for this pack)

- **Bipolar plate (series) connection** - the standard method for PEMFC (proton exchange membrane fuel cell), PAFC (phosphoric acid fuel cell), planar SOFC (solid oxide fuel cell): cells are stacked face

to face; each bipolar plate is electrically conductive and connects the anode of one cell to the cathode of the next over the whole area (short current path, low resistance). Flow channels on one face carry hydrogen to the anode, on the other face air to the cathode. Stack voltage = number of cells × cell voltage; current is the same through all cells.

- **Manifolds**: fuel, air and coolant are distributed to all cells in **parallel** through internal manifolds (holes through the plates) or external manifolds; gaskets/seals prevent leaks.
- **End plates and current collectors**: thick end plates with tie rods compress the stack (good contact, sealing); current collectors at each end take the current out.
- **Cooling plates** every few cells remove heat (air or water).
- **Edge collection / planar side-by-side** connection: for small or low-power cells (e.g. portable DMFC), cells are placed side by side and connected edge to edge with wires/strips (longer current path, more resistance).
- **Tubular SOFC**: tubes connected by ceramic interconnect strips in series and parallel bundles.

Layer structure of a DMFC



DMFC layer structure (drawn for this pack)

Layer	Material and role
Anode flow-field (end/bipolar) plate	Graphite or metal; channels feed the methanol-water solution and remove CO_2; collects current
Anode gas diffusion (backing) layer	Carbon cloth/paper, partly PTFE-treated; spreads liquid fuel evenly, lets CO_2 bubbles escape, conducts electrons
Anode catalyst layer	**Pt-Ru** (ruthenium helps oxidise the CO intermediate that poisons Pt), with Nafion ionomer; $CH_3OH + H_2O \rightarrow CO_2 + 6H^+ + 6e^-$
Proton exchange membrane	Nafion (perfluorosulfonic acid), ~100-200 µm; conducts H^+, separates fuel and air; methanol crossover through it is the main problem
Cathode catalyst layer	Pt on carbon; $3/2\ O_2 + 6H^+ + 6e^- \rightarrow 3H_2O$
Cathode gas diffusion layer	Hydrophobic carbon paper with microporous layer; supplies air, removes water
Cathode flow-field plate	Air channels; collects current

The catalyst layers + membrane form the **membrane electrode assembly (MEA)**. Overall: CH_3OH + 3/2 O_2 → CO_2 + $2H_2O$; E^0 = 1.21 V, practical 0.3-0.5 V. Operates at 60-120 °C with 1-2 M methanol. Problems: slow anode kinetics, methanol crossover (mixed potential at cathode), high Pt-Ru loading, CO_2 management.

> **Memory trick:** Stack = **series by bipolar plates, gases in parallel by manifolds**. DMFC sandwich: **"Plate, Diffusion, Pt-Ru, Nafion, Pt, Diffusion, Plate"**.

A11 Phosphoric acid fuel cell (PAFC): electrolyte properties and functions, electrodes and catalyst Must study Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2022 Q4(b)** [13] What are the properties and functions of phosphoric acid electrolytes in Phosphoric Acid Fuel Cells (PAFCs)? Provide a brief explanation of the electrode and catalyst in PAFCs.

★ REMEMBER IT

- Phosphoric acid electrolyte: held in a SiC matrix; stable, tolerates CO_2 and up to ~1 % CO; conducts H^+; low volatility above 150 °C. Electrodes: carbon paper with PTFE-bonded Pt on carbon catalyst.

Before you start: the first commercial fuel cell

The PAFC (phosphoric acid fuel cell) uses concentrated **phosphoric acid** as electrolyte, held in a thin porous sheet, and works at about 200 °C. It was the first fuel cell sold commercially (200 kW units for buildings and hospitals, e.g. the UTC PC25). The ion that moves is the proton H^+, like in a PEMFC (proton exchange membrane fuel cell).

Properties of phosphoric acid electrolyte

- **Concentrated (~100 %) H_3PO_4**, used at 150-220 °C; good **proton conductor** at these temperatures.
- **Thermal, chemical and electrochemical stability**: does not decompose, stable within the cell potential window.
- **Low volatility** above 150 °C (less evaporation loss than other acids).
- **Tolerant to CO_2** in fuel and air (unlike alkaline electrolyte, which forms carbonates); tolerates ~1 % CO at 200 °C.
- At low temperature it is a poor ionic conductor and CO poisons Pt; solidifies at 42 °C, so stack must be kept warm.
- Strong adsorption of phosphate anions on Pt slows the oxygen reduction; corrosive (needs graphite components).

Functions of the electrolyte

1. Conducts protons from anode to cathode (completes the internal circuit).
2. Separates the fuel and oxidant gases (with the matrix acting as a gas barrier).
3. Electronic insulator between the electrodes, forcing electrons through the external circuit.
4. Provides the ionic contact at the three-phase boundary in the electrodes.

The acid is held by capillary action in a **matrix of silicon carbide (SiC) particles bonded with PTFE (polytetrafluoroethylene (Teflon))** (~0.1-0.2 mm thick); an acid reservoir in the porous plates replenishes evaporation losses.

Electrodes and catalyst in PAFCs

- **Structure**: gas-diffusion electrodes - a **carbon paper** substrate (backing, ~90 % porosity) that supports a thin catalyst layer of **Pt (or Pt alloy) supported on high-surface-area carbon black**,

bonded with **PTFE** (~30-50 %) to keep pores hydrophobic for gas access while the electrolyte wets the catalyst.

- **Anode**: Pt/C, loading ~0.1-0.25 mg Pt/cm^2; $H_2 \rightarrow 2H^+ + 2e^-$.
- **Cathode**: Pt or Pt alloys (Pt-Cr-Co, Pt-Ni) on carbon, ~0.5 mg/cm^2, more active and stable for oxygen reduction: $½O_2 + 2H^+ + 2e^- \rightarrow H_2O$.
- Carbon supports: graphitised carbon black for corrosion resistance at high potential and temperature.
- Issues: Pt particle sintering and carbon corrosion reduce life; Pt cost.

PAFC features: efficiency ~40 % electrical, ~85 % with CHP; reliable, long life (> 40,000 h); but expensive Pt, low power density, slow start.

> **Memory trick:** PAFC = **"~100 % H_3PO_4 in SiC-PTFE, 200 °C, Pt on carbon on carbon paper, H^+ carries"**.

A2 Crystalline silicon structures (mono-, multi-, amorphous); diffusion and drift device concepts

Must study | Very likely · 100% | Easy

PAST-PAPER QUESTIONS

- **2022 Q1(b)** [12] Explain the various structures of crystalline silicon materials used in solar cells. Briefly explain the different structures of a solar cell based on the diffusion and drift device concepts.

★ REMEMBER IT

- Silicon types: **mono** (one crystal, best, ~24 %), **multi/poly** (many grains, cheaper, ~20 %), **amorphous** (no order, thin film, ~10 %, degrades).
- **Diffusion** device: wafer cells; carriers wander (diffuse) to the junction because the absorber is thick. **Drift** device: thin-film p-i-n cells; a built-in field across the intrinsic layer sweeps (drifts) carriers out.

Before you start: order in the crystal and how carriers travel

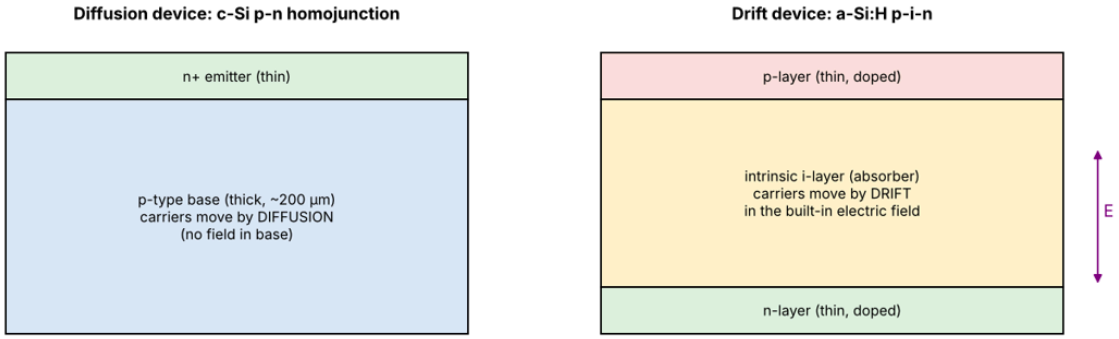
In a perfect crystal, atoms sit in a regular repeating pattern (lattice). Silicon for solar cells comes in three structures depending on how perfect this order is: **one crystal** (monocrystalline), **many small crystals** (multicrystalline/polycrystalline), or **no long-range order** (amorphous).

After light creates an electron-hole pair, the carriers must reach the junction. They can either wander randomly (**diffusion**, slow but fine if the material is very pure) or be pushed by an electric field (**drift**, needed if the material has many defects so carriers die quickly).

Structures of crystalline silicon materials

Structure	Description	Production	Solar cell efficiency
Monocrystalline (c-Si / sc-Si)	Whole wafer is one crystal; continuous lattice, no grain boundaries; uniform dark look	Czochralski (Cz) pulling from melt or float-zone (FZ) refining; wafers sawn from ingots	Highest: 20-26 %; most expensive
Multicrystalline (mc-Si / poly-Si)	Many grains (mm-cm size) with different orientations; grain boundaries act as recombination centres; bluish speckled look	Casting / directional solidification of molten Si in crucibles; also ribbon Si	16-22 %; cheaper
Microcrystalline / nanocrystalline (µc-Si:H)	Tiny crystallites embedded in amorphous tissue	PECVD (plasma-enhanced chemical vapour deposition) at high hydrogen dilution	Used in thin-film tandems
Amorphous (a-Si:H)	No long-range order; dangling bonds passivated by hydrogen; high defect density, short diffusion length; direct-like absorption (strong), $E_g \approx 1.7$ eV	PECVD from silane at ~200 °C on glass/foil	6-10 %; cheap, flexible, light-induced degradation

Diffusion and drift device concepts



Diffusion device (p-n) vs drift device (p-i-n), drawn for this pack

	Diffusion device	Drift device
Structure	p-n (homo)junction: thin heavily doped n^+ emitter on thick p-type base (~150-200 µm), back-surface field	p-i-n: thin doped p and n layers with a thick intrinsic (undoped) absorber layer between them
Where light is absorbed	Mostly in the field-free base	In the intrinsic layer, which has a strong built-in electric field across it
Carrier transport	Minority carriers (electrons in p-base) move to the junction by **diffusion** (random motion from high to low concentration)	Carriers are swept by **drift** in the electric field to the p and n layers
Requirement	Long minority-carrier diffusion length (L > base thickness): high-quality material	Works with short diffusion length (defect-rich material); i-layer kept thin (~300 nm) so field is strong
Examples	Crystalline Si cells (mono and multi), GaAs	a-Si:H thin-film cells; also n-i-p configuration on substrates

In short: if carriers live long enough, they can simply diffuse to the junction (diffusion device). In defective amorphous silicon they would recombine first, so the absorber is placed inside the electric field of a p-i-n junction (drift device). Heterojunction (HIT/SHJ) cells combine c-Si absorber with thin a-Si:H layers.

> **Memory trick: Mono = one crystal, Multi = many grains, Amorphous = no order. Diffusion for pure (c-Si), Drift for dirty (a-Si).**

A5 Chalcopyrite materials (CIS, CIGS) and the CdTe cell: layer structure and working principle

Must study · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2022 Q2(b)** [12] What are the common combinations of elements used to form chalcopyrite materials? Provide a brief explanation of the layer structure of a typical CdTe solar cell and its working principle.

★ REMEMBER IT

- Chalcopyrite = I-III-VI_2: $Cu(In,Ga)(Se,S)_2$ = CIS / CIGS (copper indium gallium (di)selenide, $Cu(In,Ga)Se_2$). CdTe cell stack (superstrate, light enters through glass): **glass → TCO (front contact) → CdS (n, window) → CdTe (p, absorber) → back contact**.

Before you start: compound semiconductors for thin films

Besides silicon, some compounds absorb sunlight very strongly (direct band gap), so a layer only 1-3 micrometres thick (50 times thinner than a Si wafer) is enough. Two famous thin-film families: **chalcopyrites** (copper-indium-gallium-selenide, CIGS) and **cadmium telluride (CdTe)**. A **heterojunction** is a p-n junction made of two different materials (e.g. n-CdS and p-CdTe).

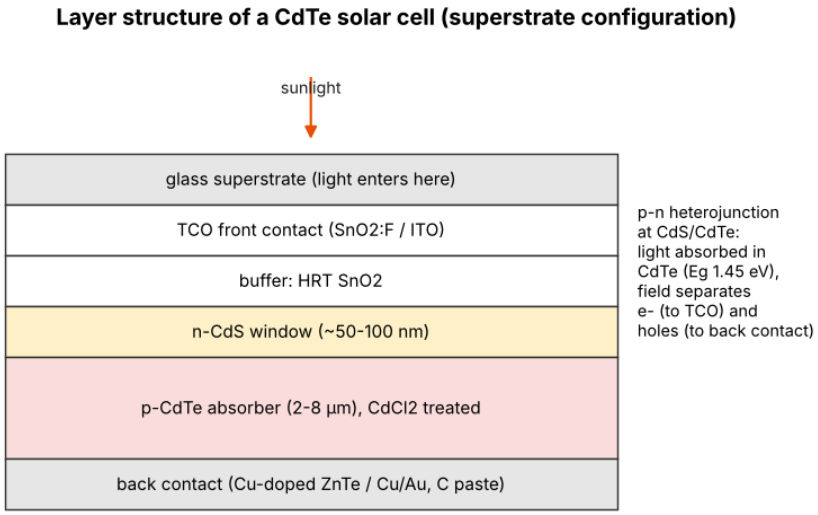
Common combinations of elements in chalcopyrite materials

Chalcopyrites are **I-III-VI_2** compounds (named after the mineral $CuFeS_2$) with a tetragonal structure derived from zinc-blende: group I = **Cu** (or Ag), group III = **In, Ga** (or Al), group VI = **Se, S** (or Te).

Compound	Band gap (eV)
$CuInSe_2$ (CIS)	1.0
$CuGaSe_2$ (CGS)	1.7
$CuInS_2$	1.5
$CuGaS_2$	2.4
$Cu(In,Ga)Se_2$ (**CIGS**, Ga/(In+Ga) ≈ 0.3)	1.1-1.2 (tunable)
$Cu(In,Ga)(Se,S)_2$ (CIGSSe)	tunable
$AgInSe_2$, $CuAlSe_2$	less common

Replacing In by Ga or Se by S raises the band gap, so the gap can be tuned (graded). Related In/Ga-free kesterite: $Cu_2ZnSn(S,Se)_4$ (CZTS). CIGS cells reach ~23 % efficiency.

Layer structure of a typical CdTe cell



CdTe solar cell layer structure, superstrate configuration (drawn for this pack)

Layer (light enters from glass)	Material and role
Glass superstrate	Mechanical support; light enters through it (superstrate configuration)
Front contact: TCO (transparent conducting oxide)	SnO_2:F or ITO (indium tin oxide); transparent and conductive, collects electrons
High-resistive transparent buffer	Intrinsic SnO_2 or ZnO; allows a thinner CdS and prevents shunts
n-type window: CdS	~50-100 nm, E_g = 2.4 eV, forms the heterojunction; kept thin to let blue light through (newer cells use CdSe/MgZnO)
p-type absorber: CdTe	2-8 µm, direct E_g = 1.45 eV (ideal for sunlight), absorption coefficient > 10^4 cm^{-1}; deposited by close-space sublimation; **$CdCl_2$ activation treatment** recrystallises grains and passivates grain boundaries
Back contact	Cu-doped ZnTe, Cu/Au, or graphite paste + metal; low-barrier ohmic contact to p-CdTe (difficult because of CdTe's high work function)

Working principle: sunlight passes through glass, TCO and the thin CdS window and is absorbed in the CdTe, creating electron-hole pairs. The built-in electric field of the n-CdS/p-CdTe heterojunction (depletion region extends mainly into the CdTe) separates them: electrons drift to the CdS and TCO front contact, holes go to the back contact; current flows through the external load. Efficiency up to ~22 % (First Solar); low cost per watt, but Cd toxicity and Te scarcity are concerns.

> **Memory trick:** Chalcopyrite = **I-III-VI_2: Cu-(In,Ga)-$(Se,S)_2$**. CdTe stack: **"Glass, TCO, Buffer, CdS, CdTe, Back contact"**.

A9 Role of PTFE in alkaline fuel cells; functions of bipolar plates in PEMFCs

Must study | Very likely · 100% | Easy

PAST-PAPER QUESTIONS

- **2022 Q3(c)** [12] What is the role of PTFE (polytetrafluoroethylene (Teflon)) in Alkaline Fuel Cells (AFCs)? What are the functions of bipolar plates in Proton Exchange Membrane Fuel Cells (PEMFCs)?

★ REMEMBER IT

- PTFE (Teflon) in AFC (alkaline fuel cell) electrodes makes them **water-repellent**: keeps pores open for gas so the liquid KOH does not flood them, and binds the catalyst particles.
- Bipolar plate jobs (**"CDS-HS"**): Conduct electrons between cells, Distribute gases via flow channels, Separate fuel and oxidant, Heat removal, Support the stack (mechanical).

Before you start: water: friend and enemy

Electrode reactions happen only where three things meet: the **gas**, the **electrolyte** (to carry ions) and the **catalyst on carbon** (to carry electrons) - the **three-phase boundary**. If liquid floods the pores, gas cannot reach the catalyst; if it is too dry, ions cannot move.

PTFE (polytetrafluoroethylene, Teflon) is a water-repelling plastic, used to control this.

Role of PTFE in alkaline fuel cells (AFC)

AFCs use a liquid KOH electrolyte (30-45 %) and porous gas-diffusion electrodes (Ni, Ag or Pt on carbon). PTFE is mixed into and coated on these electrodes:

1. **Hydrophobic (water-repellent) agent**: keeps the liquid KOH electrolyte and product water from **flooding** the electrode pores, so gas channels stay open for H_2/O_2.
2. **Creates and stabilises the three-phase boundary**: hydrophilic catalyst regions wetted by electrolyte sit next to hydrophobic PTFE gas pores - maximum active area.
3. **Binder**: holds catalyst and carbon particles together and to the backing (nickel mesh), giving mechanical strength.
4. **Gas-side barrier layer**: a PTFE layer on the gas side stops electrolyte leaking out through the electrode (weeping) while letting gas in.
5. Chemically stable in strong alkali and at operating temperature; controls porosity of the electrode.

Functions of bipolar plates in PEMFCs

1. **Electrical connection**: connect the anode of one cell to the cathode of the next, putting cells in series and conducting current through the stack (need high conductivity).
2. **Gas distribution**: flow-field channels (serpentine, parallel, interdigitated) distribute hydrogen and air evenly over the whole active area of each MEA (membrane electrode assembly).
3. **Separation of reactants**: impermeable plate keeps fuel and oxidant of adjacent cells apart (prevents mixing and combustion).
4. **Water management**: carry away product water from the cathode, preventing flooding.
5. **Heat management**: conduct heat out; often contain coolant channels.
6. **Mechanical support**: give rigidity to the stack, support the thin MEA under compression.

Requirements and materials: high electrical and thermal conductivity, corrosion resistance in the acidic humid environment, gas impermeability, low weight and cost, easy manufacture. Materials: machined graphite, expanded graphite/polymer composites, coated stainless steel or titanium. Bipolar plates are ~80 % of stack weight and a large share of cost.

> **Memory trick:** PTFE = **"keep the pores dry, the electrolyte in, the catalyst together"**. Bipolar plate = **"Connect, Distribute, Separate, Drain, Cool, Support"**.

A12 Solid oxide fuel cell (SOFC): favourable characteristics, electrolyte types

Must study | Very likely · 100% | Easy

PAST-PAPER QUESTIONS

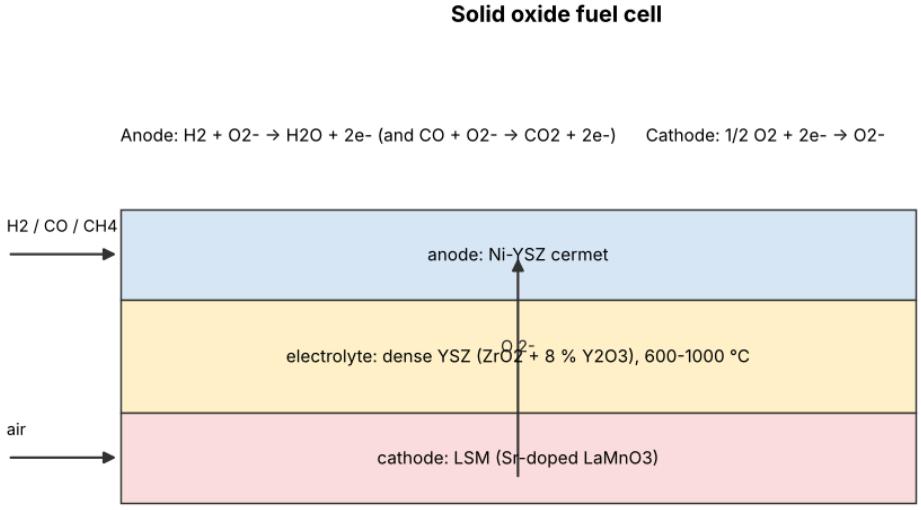
- **2022 Q4(c)** [12] Explain the favorable characteristics of a Solid Oxide Fuel Cell (SOFC). Explain the types of electrolytes used in this fuel cell.

★ REMEMBER IT

- SOFC (solid oxide fuel cell) favourites: all solid (no liquid to leak), high temperature → no precious metal catalyst, internal reforming, fuel flexible (H_2, CO, CH_4), hot exhaust for CHP or gas turbine. Electrolytes: YSZ (yttria-stabilised zirconia, most common), scandia-stabilised zirconia, doped ceria (GDC), LSGM (strontium- and magnesium-doped lanthanum gallate).

Before you start: an all-ceramic fuel cell

In a **solid oxide fuel cell** every part is solid - mostly ceramics. At high temperature (600-1000 °C) certain oxides let **oxide ions O^{2-}** hop through vacancies in their crystal lattice; that is the electrolyte. Oxygen ions travel from the cathode (air side) to the anode, where they burn the fuel electrochemically.



Solid oxide fuel cell (drawn for this pack)

Anode: $H_2 + O^{2-} \rightarrow H_2O + 2e^-$; $CO + O^{2-} \rightarrow CO_2 + 2e^-$ Cathode: $\frac{1}{2}O_2 + 2e^- \rightarrow O^{2-}$

Favourable characteristics of SOFC

1. **High efficiency**: 45-60 % electrical; up to ~70 % in hybrid SOFC-gas turbine systems and 85-90 % with CHP.
2. **Fuel flexibility**: H_2, CO, natural gas, biogas, propane, even ammonia; CO is a fuel; **internal reforming** of hydrocarbons at operating temperature.

3. **No precious-metal catalysts** (Ni anode, perovskite cathode) - fast kinetics at high temperature.
4. **All-solid construction**: no liquid electrolyte, no corrosion/leakage or electrolyte management; flexible geometries (planar, tubular).
5. **High-quality waste heat** for cogeneration and bottoming cycles.
6. Low emissions (no NO_x formation from combustion), quiet, modular; reversible operation as an electrolyser (SOEC).
7. Tolerant to impurities better than low-temperature cells.

(Drawbacks: slow start-up, thermal cycling stresses, sealing, expensive high-temperature materials; research aims at intermediate temperatures 500-750 °C.)

Types of electrolytes used in SOFC

Electrolyte	Description	Temperature
YSZ (yttria-stabilised zirconia): yttria-stabilised zirconia (ZrO_2 + 8 mol % Y_2O_3)	Most common; Y^{3+} replacing Zr^{4+} creates oxygen vacancies; stabilises the cubic phase; purely ionic, stable in oxidising and reducing gases, cheap	800-1000 °C (needs thin films for lower T)
ScSZ (scandia-stabilised zirconia): scandia-stabilised zirconia	Higher conductivity than YSZ; costly, ageing	700-850 °C
Doped ceria: GDC (Gd-doped CeO_2), SDC (Sm-doped)	High ionic conductivity at lower temperature; but Ce^{4+} reduces to Ce^{3+} in fuel, giving electronic conduction (internal short)	500-700 °C (intermediate-temperature SOFC)
LSGM: Sr- and Mg-doped lanthanum gallate (perovskite)	High oxide-ion conductivity, stable; reacts with Ni anodes, Ga volatility	600-800 °C
Proton-conducting perovskites (doped $BaCeO_3$, $BaZrO_3$)	Conduct H^+ instead of O^{2-} (PCFC); water formed at cathode, no fuel dilution	400-700 °C
Bismuth oxide-based, apatites	Very high conductivity but unstable in reducing gases	research

Typical cell: Ni-YSZ cermet anode / YSZ electrolyte / LSM (or LSCF (lanthanum strontium cobalt ferrite) with GDC barrier) cathode; interconnects of doped $LaCrO_3$ or ferritic steel.

> **Memory trick:** SOFC = **"all solid, O^{2-} moves, hot, eats any fuel, no platinum"**. Electrolytes = **"YSZ, ScSZ, Ceria, LSGM, proton perovskites"**.

A4 A third-generation thin-film cell: dye-sensitised solar cell (working principle) Must study Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2022 Q2(a)** [10] Mention a third generation solar cell belonging to the group of thin film solar cells. Explain the working principle of such a solar cell with a suitable diagram.

UNDERSTAND **Read this to understand; you do not need to write it**

Dye-sensitised solar cell (Grätzel cell) copies a leaf. A dye coated on porous TiO_2 absorbs light, its excited electron jumps into the TiO_2 and flows to the front glass (FTO) and the external circuit. The dye is then refilled with an electron from an iodide/triiodide electrolyte, which is itself regenerated at the platinum counter electrode. Nothing is consumed: it is a cycle.

★ REMEMBER IT

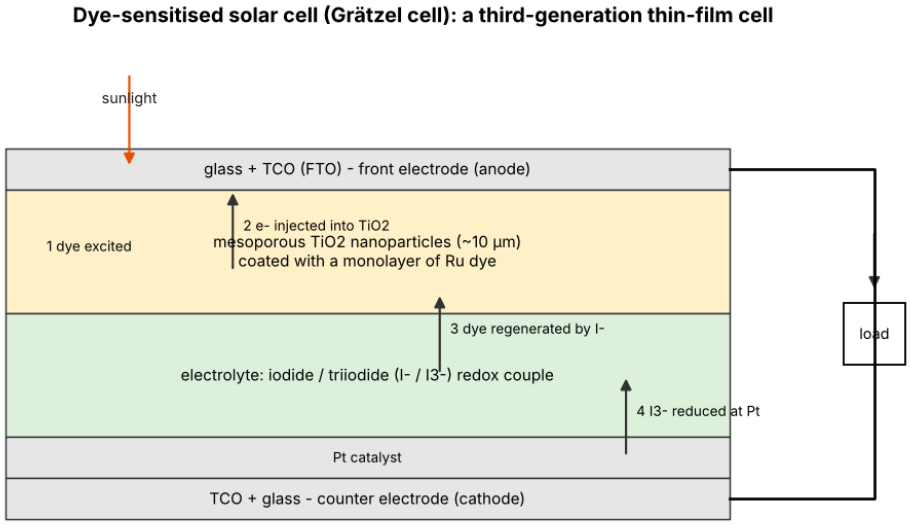
- DSSC (dye-sensitised solar cell) path: **light → dye excited → TiO_2 → FTO (fluorine-doped tin oxide) → load → Pt counter electrode → I_3^- → I^- → dye**.

Before you start: generations of solar cells

1st generation: crystalline silicon wafers. **2nd generation**: thin films (a-Si, CdTe, CIGS (copper indium gallium (di)selenide, $Cu(In,Ga)Se_2$)) - less material, cheaper. **3rd generation**: new concepts that aim for very low cost and/or efficiency beyond the single-junction limit - dye-sensitised, organic, perovskite and quantum-dot cells (thin films), plus multijunction, spectral conversion, hot-carrier concepts.

The **dye-sensitised solar cell** (DSSC, Grätzel cell, 1991) copies photosynthesis: a dye absorbs the light (like chlorophyll) and other materials carry the charges.

A third-generation thin-film cell: the dye-sensitised solar cell (DSSC)



Structure and working principle of a dye-sensitised solar cell (drawn for this pack)

Structure (sandwich): (1) glass coated with a transparent conducting oxide (FTO) as the front electrode; (2) a ~10 µm **mesoporous TiO_2** layer of nanoparticles (huge surface area) coated with a monolayer of **dye** molecules (ruthenium complexes such as N719, or organic dyes); (3) a liquid **electrolyte** containing the iodide/triiodide (I^-/I_3^-) redox couple; (4) a **counter electrode** of FTO glass with a thin platinum (or carbon) catalyst; sealed at the edges.

Working principle:

1. **Light absorption**: a photon excites the dye from its ground state S to an excited state S* (S + hv → S*).
2. **Electron injection**: the excited dye injects an electron into the conduction band of TiO_2 within femtoseconds ($S^* \rightarrow S^+ + e^-_{TiO2}$).
3. **Transport**: electrons diffuse through the TiO_2 network to the FTO and flow through the external load (doing work) to the counter electrode.
4. **Dye regeneration**: the oxidised dye takes an electron from iodide in the electrolyte: $2S^+ + 3I^- \rightarrow 2S + I_3^-$.
5. **Electrolyte regeneration**: triiodide diffuses to the Pt counter electrode and is reduced: $I_3^- + 2e^- \rightarrow 3I^-$. The cycle is complete; no net chemical change.
6. The voltage = difference between the TiO_2 Fermi level and the redox potential of the electrolyte (~0.7-0.8 V).

Features: cheap materials and simple printing processes, works well in diffuse/low light, semi-transparent and colourful (building windows), efficiency ~11-14 %. **Drawbacks:** liquid electrolyte can leak/evaporate, long-term stability, lower efficiency than Si. (Perovskite cells, which grew from DSSC research, reach > 25 %; organic cells use a donor-acceptor bulk heterojunction.)

> **Memory trick:** DSSC cycle: **"Dye excites, Injects to TiO_2, electron does work, Iodide repairs dye, Platinum repairs iodide"**.

A7 Fuel cell: definition, difference from a battery, working principle Must study Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2022 Q3(a)** [10] What is a fuel cell and how does it differ from a conventional battery? Also briefly explain the working principle of a fuel cell.

LEARN IT FROM ZERO

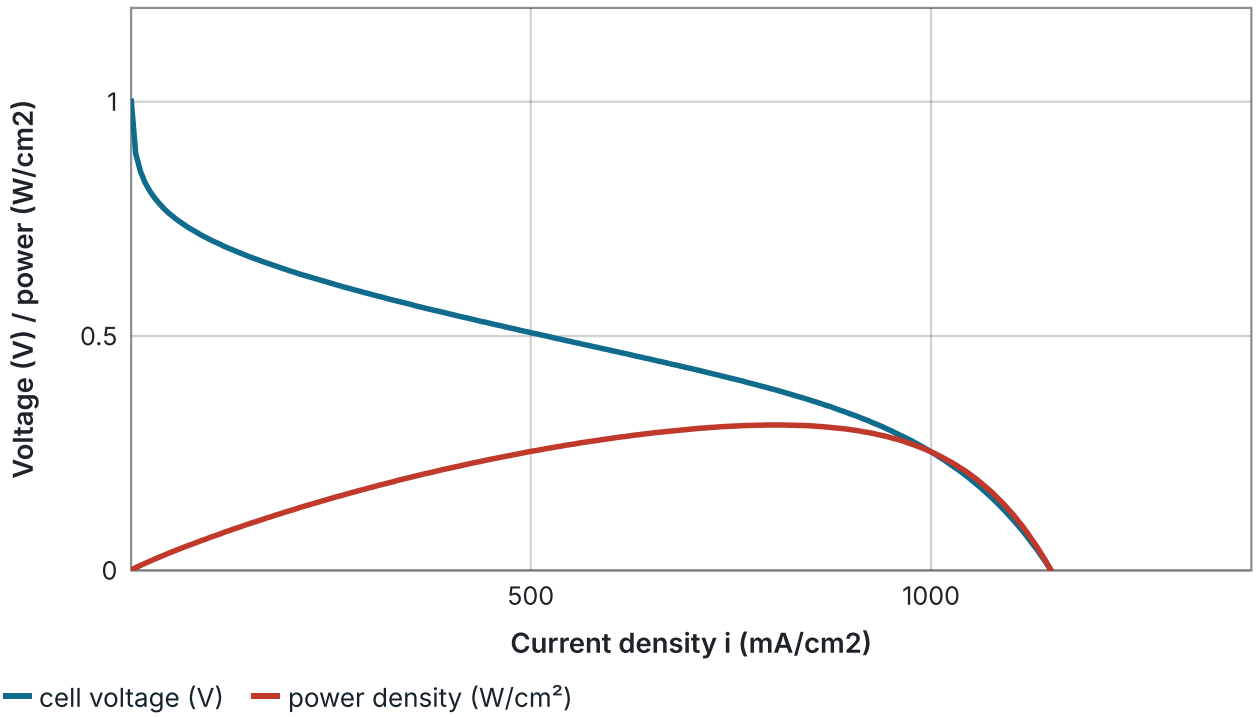
Fuel cells, from zero

A fuel cell is a battery that never runs down because you keep feeding it fuel. Hydrogen goes to the **anode**, where a catalyst splits it into protons and electrons. The electrolyte lets only ions through, so the electrons are forced round the outside circuit (that is the electric current). At the **cathode** the electrons, protons and oxygen from air join to make water.

A **battery** stores its chemicals inside and is used up (or must be recharged); a fuel cell is an **energy converter**: its electrodes are not consumed, and it runs as long as fuel and air flow. Real cell voltage is below the ideal 1.23 V because of three losses shown in the graph: **activation** (slow reactions, steep drop at low current), **ohmic** (resistance of electrolyte and contacts, straight slope) and **concentration** (gas cannot reach the catalyst fast enough at high current).

INTERACTIVE **Fuel cell polarisation curve**

Move the sliders and watch the curve change.



Region 1 (left): activation loss. Region 2 (middle, straight): ohmic loss. Region 3 (right, plunge):

concentration loss.

Before you start: a battery that never runs flat

A **fuel cell** makes electricity from a chemical reaction - usually hydrogen with oxygen from air, giving only water and heat - without burning. It is like a battery, but the chemicals are fed in continuously from outside, so it keeps working as long as fuel is supplied.

Because there is no combustion and no heat-engine cycle, it is not limited by the Carnot efficiency, and it is quiet with no moving parts.

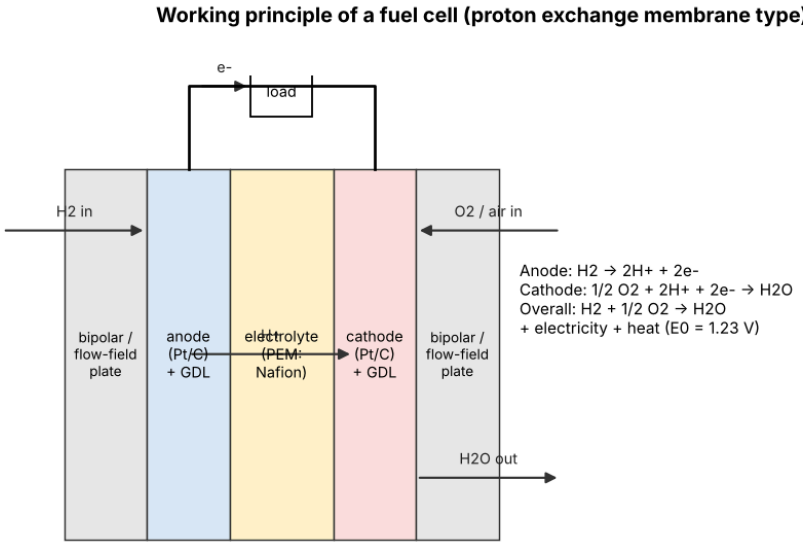
What is a fuel cell

A **fuel cell** is an electrochemical energy-conversion device that continuously converts the chemical energy of a fuel (hydrogen, methanol, natural gas) and an oxidant (oxygen/air) directly into electrical energy (plus heat and reaction products), as long as fuel and oxidant are supplied. Each cell has an anode, a cathode and an electrolyte between them.

Fuel cell vs conventional battery

Fuel cell	Battery
Energy **converter**: reactants supplied continuously from external tanks	Energy **storage** device: reactants stored inside the cell
Runs as long as fuel is supplied; never "discharged"	Runs down when reactants are used; must be recharged or replaced
Refuelled in minutes	Recharging takes hours (secondary) or impossible (primary)
Power (cell size) and energy (tank size) are sized independently	Power and energy both fixed by the cell
Electrodes are catalytic and not consumed	Electrode materials are consumed/converted
Needs balance of plant (fuel supply, humidifier, cooling, pumps)	Self-contained
Higher energy density with stored H_2; product water	Lower energy density; no emissions at use

Working principle



Working principle of a hydrogen fuel cell (PEM type), drawn for this pack

1. Hydrogen is fed to the **anode**. On the platinum catalyst it splits into protons and electrons: $H_2 \rightarrow 2H^+ + 2e^-$.
2. The **electrolyte** (e.g. a proton exchange membrane) lets only ions pass; electrons cannot cross it.
3. Electrons therefore travel through the **external circuit** to the cathode, delivering electrical power to the load.
4. Protons move through the electrolyte to the **cathode**, where oxygen from air combines with them and the electrons: $½O_2 + 2H^+ + 2e^- \rightarrow H_2O$.
5. Overall: $H_2 + ½O_2 \rightarrow H_2O$ + electricity + heat. Ideal voltage $E^0 = -\Delta G/(2F) = 1.23$ V at 25 °C; in practice ~0.6-0.8 V per cell because of activation, ohmic and concentration losses, so cells are stacked (A8).

In other types the mobile ion differs: OH^- in AFC (cathode to anode), CO_3^{2-} in MCFC (molten carbonate fuel cell), O^{2-} in SOFC (solid oxide fuel cell), H^+ in PEMFC (proton exchange membrane fuel cell)/PAFC/DMFC.

> **Memory trick: "Fuel cell converts, battery stores"**. Anode = fuel side, **A**node loses electrons (oxidation). $H_2 + ½O_2 \rightarrow H_2O$, 1.23 V.

A10 Molten carbonate fuel cell (MCFC): construction, principle, features

Must study | Very likely · 100% | Easy

PAST-PAPER QUESTIONS

- **2022 Q4(a)** [10] Briefly explain the construction and operating principle of Molten Carbonate Fuel Cells (MCFCs). Also what are the general features of MCFCs?

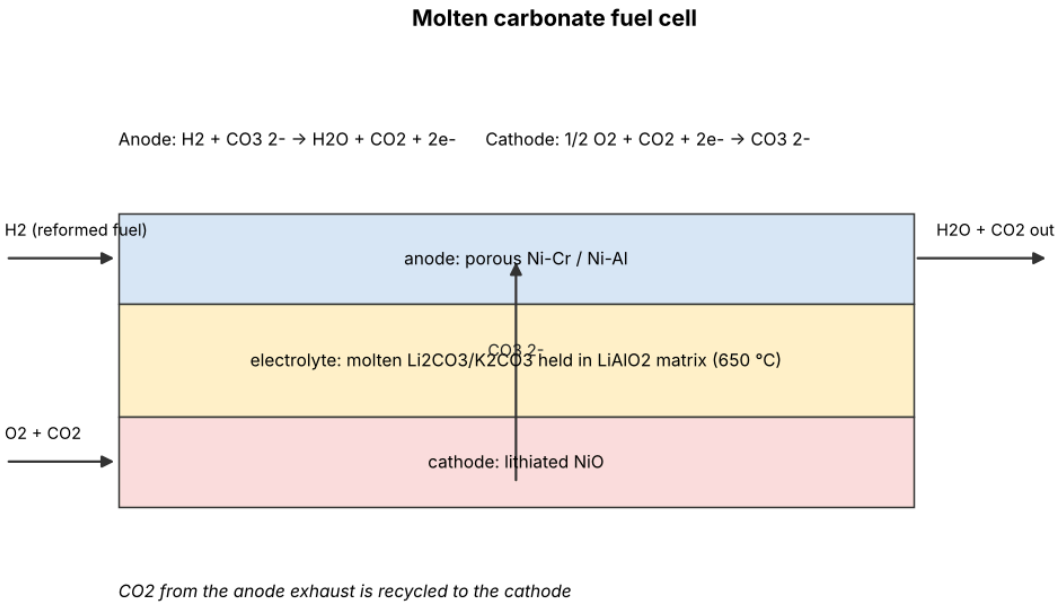
★ REMEMBER IT

- Fuel cell table by temperature: **PEMFC** (proton exchange membrane fuel cell) 80 °C (polymer, H^+), **AFC** (alkaline fuel cell) 70–200 °C (KOH, OH^-), **PAFC** (phosphoric acid fuel cell) ~200 °C (H_3PO_4, H^+), **MCFC** (molten carbonate fuel cell) ~650 °C (molten Li/K carbonate, CO_3^{2-}), **SOFC** (solid oxide fuel cell) 800–1000 °C (ceramic YSZ (yttria-stabilised zirconia), O^{2-}).
- MCFC: CO_2 must be fed to the cathode (it carries CO_3^{2-} across); hot enough for internal reforming of natural gas, no platinum needed.

Before you start: a hot fuel cell with molten salt

The MCFC uses a **molten salt** (a mixture of lithium and potassium carbonates that melts at about 500 °C) as the electrolyte, operating at about 650 °C. The ion that moves is **carbonate**, CO_3^{2-}, from cathode to anode. Because it is hot, cheap nickel works as catalyst and natural gas can be reformed inside the cell.

Construction



Molten carbonate fuel cell (drawn for this pack)

Component	Material
Electrolyte	Molten Li_2CO_3/K_2CO_3 (or Li/Na) eutectic, held by capillary forces in a porous ceramic **$LiAlO_2$ matrix** (tile ~0.5-1 mm)
Anode	Porous Ni with a few % Cr or Al (stops sintering and creep), 0.2-1.5 mm
Cathode	Porous lithiated NiO (Ni oxidised and lithiated in situ)
Separator / bipolar plates	Stainless steel (Ni-coated on anode side), with wet-seal edges; stacks of up to hundreds of cells, plants 300 kW-MW scale

Operating principle (about 650 °C)

$$\text{Anode: } H_2 + CO_3^{2-} \rightarrow H_2O + CO_2 + 2e^- \quad (\text{also } CO + CO_3^{2-} \rightarrow 2CO_2 + 2e^-)$$

$$\text{Cathode: } \tfrac{1}{2}O_2 + CO_2 + 2e^- \rightarrow CO_3^{2-} \quad \text{Overall: } H_2 + \tfrac{1}{2}O_2 + CO_2(\text{cathode}) \rightarrow H_2O + CO_2(\text{anode})$$

Air plus CO_2 is fed to the cathode, where carbonate ions form; they migrate through the molten electrolyte to the anode and react with hydrogen (from reformed natural gas/biogas), releasing electrons to the external circuit, water and CO_2. The CO_2 produced at the anode is **recycled** to the cathode (by burning the anode exhaust and mixing with cathode air).

General features of MCFCs

- High operating temperature (600-700 °C): fast kinetics, **no precious-metal catalyst** (Ni).
- **Fuel flexibility**: natural gas, biogas, coal gas; CO is a fuel, not a poison; **internal reforming** of methane possible.
- High electrical efficiency (45-55 %), > 80 % with combined heat and power or bottoming cycle; useful high-grade waste heat.
- Suitable for stationary MW-scale power plants; can also concentrate CO_2 (carbon capture).
- Drawbacks: corrosive molten electrolyte, electrolyte loss and NiO cathode dissolution limit life; need CO_2 recycle; slow start-up (not for vehicles); sulphur sensitive; complex sealing.

> **Memory trick:** MCFC = **"Molten carbonate, 650 °C, CO_3^{2-} carries, Ni cheap, CO_2 recycled"**.

A3 High-efficiency concepts of c-Si cells; the PERL cell Must study

Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2022 Q1(c)** [13] What is meant by high-efficiency concepts of solar cells? Explain with a proper schematic diagram how the PERL (passivated emitter, rear locally-diffused (cell)) concept contributes to the overall advancement of crystalline silicon solar cell technology.

★ REMEMBER IT

- PERL = **P**assivated **E**mitter, **R**ear **L**ocally diffused. Each word fixes a loss: passivated emitter (less front recombination), inverted pyramids + AR coating (less reflection), rear oxide + local contacts only at points (less rear recombination, better rear reflection). Record ~25 %.

Before you start: pushing silicon to its limit

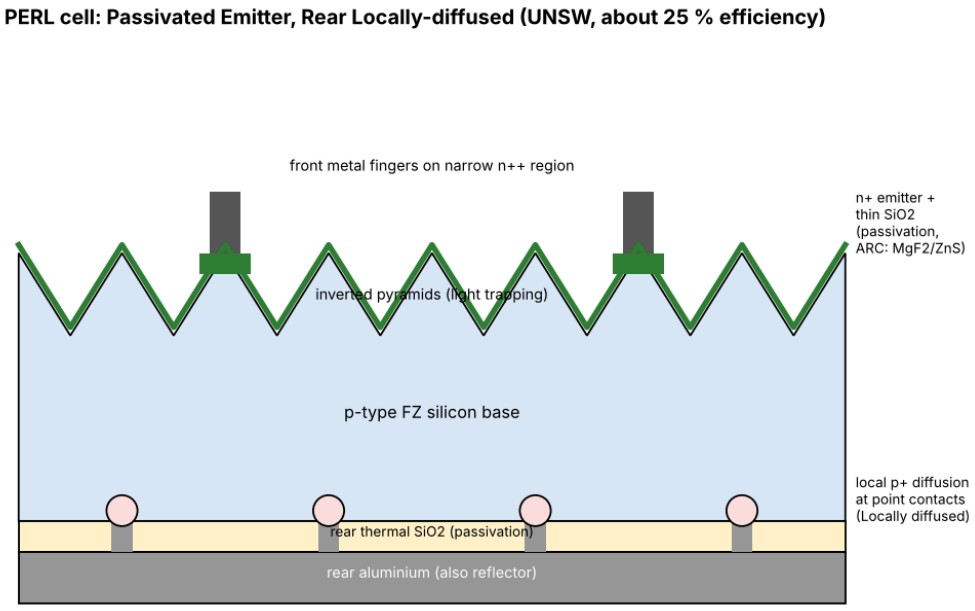
A normal industrial Si cell loses efficiency because (1) light is reflected or escapes, (2) electrons and holes recombine at the surfaces and contacts, (3) resistance in the metal grid. **High-efficiency concepts** are design tricks that attack each of these losses. The **PERL** cell from the University of New South Wales (Martin Green's group) used all of them together and held the world record (~25 %) for about 15 years.

High-efficiency concepts of solar cells

High-efficiency concepts are cell designs that approach the theoretical efficiency limit of silicon by minimising optical, recombination and resistive losses:

- **Light management**: surface texturing (random or inverted pyramids), double-layer anti-reflection coatings, rear reflectors → light trapping.
- **Surface passivation**: thermal SiO_2, Al_2O_3, SiN_x or a-Si:H layers on front and rear to reduce surface recombination.
- **Selective / lightly doped emitter**: light doping under the illuminated area (less recombination, better blue response), heavy doping only under contacts.
- **Local (point) contacts**: metal touches Si only on a small fraction of the area, reducing contact recombination; local back-surface field.
- **High-quality wafers**: float-zone or n-type Si with long carrier lifetime.
- Other structures: **PERC** (passivated emitter and rear cell) (passivated emitter and rear cell), **PERT** (passivated emitter, rear totally-diffused), **IBC** (interdigitated back contact) (interdigitated back contact, no front shading), **HIT/SHJ** (heterojunction with intrinsic thin layer), **TOPCon** (tunnel oxide passivated contact) (tunnel oxide passivated contacts).

The PERL cell and how it advanced c-Si technology



PERL cell cross-section (drawn for this pack)

PERL = Passivated Emitter, Rear Locally-diffused cell (Zhao, Wang and Green, UNSW (University of New South Wales), 1999: 24.7 %, later 25.0 %).

Feature	Purpose (which loss it reduces)
Inverted pyramids on the front (photolithography and etching)	Low reflection and strong light trapping (optical loss)
Double-layer ARC (anti-reflection coating) (MgF_2/ZnS) on a thin thermal oxide	Reflection below ~2 %; oxide passivates the front surface
Lightly diffused (passivated) n^+ emitter with heavy n^{++} diffusion only under the fingers (selective emitter)	Low emitter recombination and good blue response; low contact resistance
Narrow, high front metal fingers (Ti/Pd/Ag)	Low shading and low series resistance
Rear thermal oxide passivation	Very low rear-surface recombination; oxide + Al also form an excellent rear reflector
Rear local point contacts through holes in the oxide, with **local p^+ diffusion** (boron) under each contact	Contact recombination limited to tiny areas; local back-surface field repels minority carriers
High-lifetime float-zone p-type wafer	Long diffusion length, high V_{oc} (~700 mV)

Contribution to c-Si technology: PERL proved that silicon can exceed 25 %, showed the value of surface passivation, rear reflectors and local contacts, and became the blueprint for today's mass-produced **PERC** cells (which simplify PERL using Al_2O_3/SiN_x rear passivation and laser-opened local contacts, ~22-23 % in production), and for PERT, TOPCon and IBC designs. Its drawback was costly processing (photolithography, multiple high-temperature steps, FZ wafers), which later work made industrial.

> **Memory trick:** PERL = **P**assivated **E**mitter, **R**ear **L**ocally-diffused: **"pyramids on top, oxide all round, dots at the back"**.

A6 Multijunction cells: challenges; spectral conversion Must study

Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2022 Q2(c)** [13] Briefly explain the challenges associated with multijunction solar cells. What is spectral conversion in the context of solar cells, and how does it enhance their efficiency?

UNDERSTAND **Read this to understand; you do not need to write it**

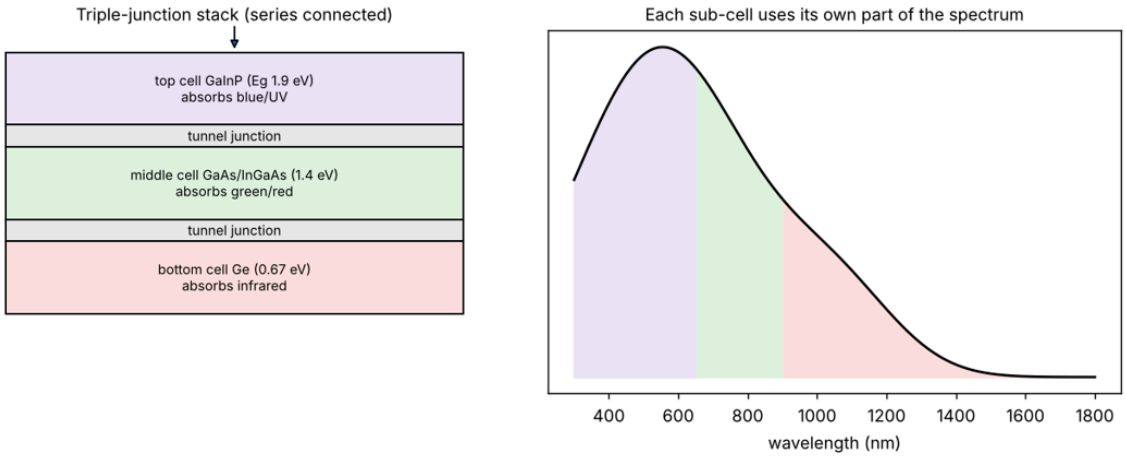
A multijunction cell stacks cells with different band gaps: the top cell takes blue light, lower cells take red and infrared. Each photon is used by a cell whose gap matches it, so less heat is wasted (efficiency > 40 %). Challenges: lattice matching between layers, current matching (series-connected cells must give the same current), tunnel junctions between cells, high cost.

Spectral conversion changes the light before it reaches a single cell: **down-conversion** splits one high-energy photon into two lower ones; **up-conversion** joins two low-energy photons into one usable one; **down-shifting** moves UV into visible light.

Before you start: beating the single-junction limit

One band gap cannot fit the whole solar spectrum: blue photons waste their extra energy as heat (thermalisation), infrared photons below the gap are not absorbed at all. Two ways out: **use several cells with different gaps** (multijunction), or **change the sunlight** to fit the cell (spectral conversion).

Multijunction solar cells and their challenges



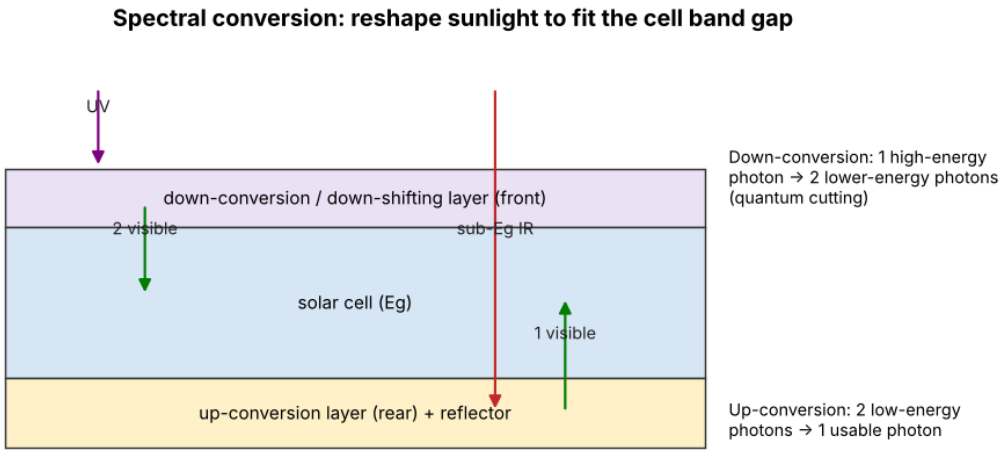
Triple-junction cell and spectrum splitting (drawn for this pack)

A **multijunction (tandem)** cell stacks sub-cells of decreasing band gap from top to bottom (e.g. GaInP 1.9 eV / GaAs 1.4 eV / Ge 0.67 eV). High-energy photons are absorbed by the top cell, lower-energy photons pass to the lower cells; thermalisation and non-absorption losses are reduced. Record efficiency > 47 % under concentration; used in space and concentrator PV. Perovskite/Si tandems reach ~34 %.

Challenges:

1. **Current matching**: in a monolithic two-terminal stack the sub-cells are in series, so the current is limited by the weakest sub-cell; band gaps and thicknesses must be designed precisely.
2. **Spectral sensitivity**: the current balance changes with the sun's spectrum (time of day, air mass, clouds), lowering real-world energy yield.
3. **Lattice matching**: epitaxial layers must have nearly the same lattice constant, otherwise dislocations cause recombination (metamorphic buffers add complexity).
4. **Tunnel junctions** between sub-cells must be transparent and low-resistance.
5. **Cost and complexity**: MOCVD (metal-organic CVD (chemical vapour deposition)) growth, expensive III-V materials and substrates (Ge, GaAs) - economical only with concentrators or in space.
6. **Thermal and mechanical mismatch**; heat removal under concentration; tracking needed for concentrators.
7. For four-terminal mechanically stacked cells: extra optical losses and wiring.

Spectral conversion and how it enhances efficiency



Spectral conversion layers on a solar cell (drawn for this pack)

Spectral conversion means modifying the incoming solar spectrum with an optically active layer (luminescent materials, rare-earth ions such as Er^{3+}, Yb^{3+}, Pr^{3+}, quantum dots, organic dyes) so that more photons have energies the cell uses efficiently. The cell itself is unchanged.

Process	Mechanism	How it raises efficiency
Down-conversion (quantum cutting)	One high-energy photon ($E > 2E_g$) is converted into two lower-energy photons (each $> E_g$), quantum efficiency up to 200 %	Reduces thermalisation loss: one UV photon gives two electron-hole pairs
Down-shifting (luminescent down-shifting)	One UV/blue photon is converted into one lower-energy photon where the cell's response is better	Improves poor blue response (e.g. CdS absorption in CdTe, surface recombination)
Up-conversion	Two (or more) sub-band-gap photons are combined into one photon with $E > E_g$ (placed at the rear)	Reduces non-absorption (transmission) loss: uses infrared light

Theoretical limits: about 39 % with ideal down-conversion and about 47 % with ideal up-conversion for a single-junction cell (vs ~33 %). Practical gains are still small due to low conversion efficiency.

> **Memory trick:** Multijunction problems = **"Current, Spectrum, Lattice, Tunnel, Cost"**. Spectral: **Down = cut one big into two small; Up = glue two small into one big**.

A1 Losses in solar cells; optical losses Must study Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2022 Q1(a)** [10] What are the different types of losses encountered in solar cells? Explain the concept of optical losses in solar cells with the proper schematic diagram.

LEARN IT FROM ZERO

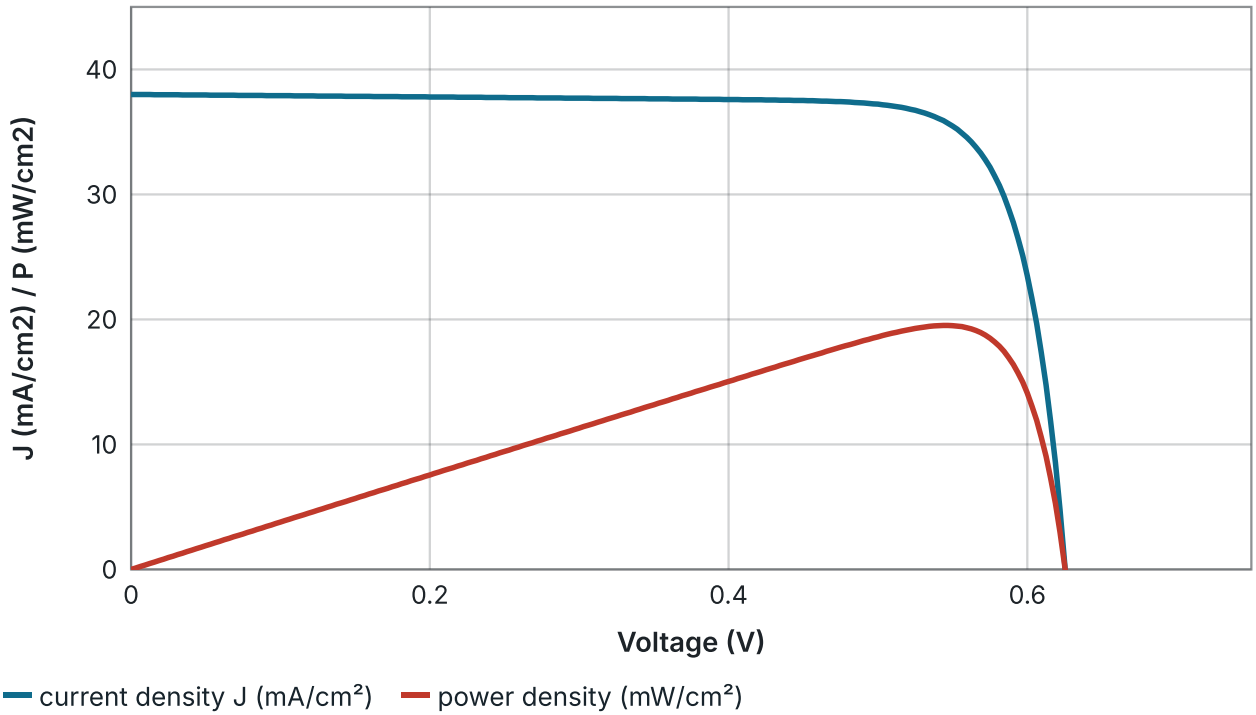
Solar cell losses, from zero

A solar cell turns light into current (from absorbed photons) at a voltage (set by the band gap). Every loss eats one of those. Losses come in three families:

- **Optical**: light never gets absorbed: reflection from the front surface, shading by the metal fingers, light passing straight through a thin cell. Fixes: anti-reflection coating, texturing (pyramids), thin fingers, back reflector.
- **Recombination (electrical)**: an electron-hole pair is made but meets again before it reaches the contacts (at defects, surfaces). Fixes: passivation, pure material.
- **Resistive**: series resistance in fingers, contacts, emitter; shunt paths that leak current. Shows up as a poor fill factor.
- **Fundamental (thermalisation and non-absorption)**: photons below the band gap pass through; photons far above it waste their extra energy as heat. This is why a single cell cannot beat about 33 % (Shockley-Queisser).

INTERACTIVE **Solar cell J-V curve: what recombination and shunt losses do**

Move the sliders and watch the curve change.



V_{oc} (ideal): **0.626** V

A low shunt resistance tilts the flat top of the curve; a higher J_0 (more recombination) lowers V_{oc}.

Both shrink the maximum power (peak of the red curve).

Before you start: how a solar cell works and where energy is lost

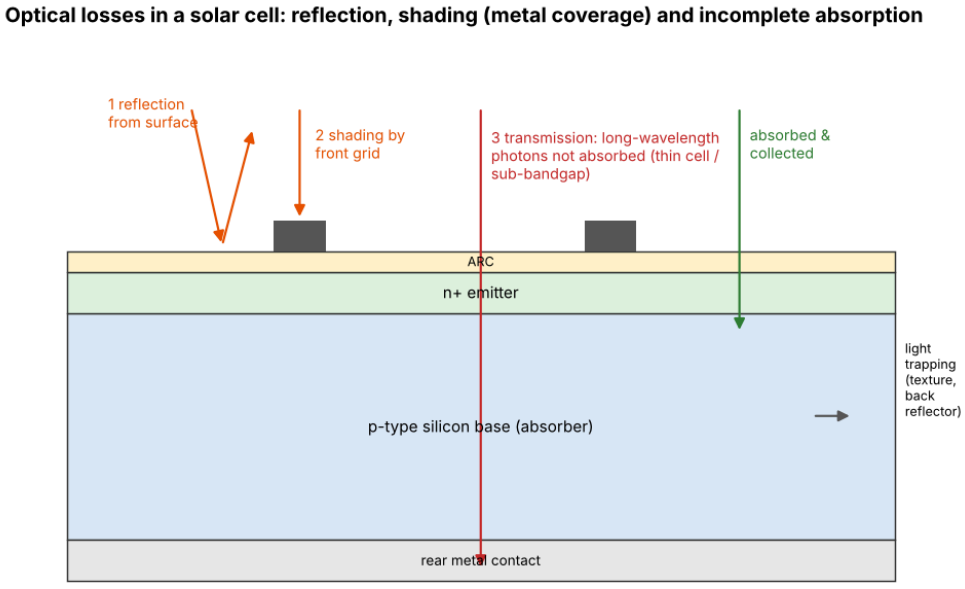
Sunlight is a stream of **photons** (energy packets). In a semiconductor such as silicon, a photon with energy above the **band gap** E_g (1.12 eV for Si) can lift an electron free, creating an **electron-hole pair**. The p-n junction's electric field pushes electrons one way and holes the other, so a current flows through the external circuit.

Every step can lose energy: light may not get in (optical loss), photons may be too weak or too strong (spectral loss), the freed carriers may recombine before reaching the contacts (recombination loss), or lose voltage in resistances (resistive loss). The best Si cells reach about 26-27 %; the theoretical limit is about 33 % (Shockley-Queisser).

Types of losses in solar cells

Loss	What happens	Remedy
Optical losses	Light does not enter or is not absorbed: reflection, shading by front contacts, transmission (incomplete absorption)	Anti-reflection coating (ARC), texturing, thin fingers, light trapping, back reflector
Spectral (fundamental) losses	**Non-absorption**: photons with $E < E_g$ pass through. **Thermalisation**: excess energy $E - E_g$ of high-energy photons lost as heat	Multijunction cells, spectral conversion (A6)
Recombination losses	Electron-hole pairs recombine (radiative, Auger, Shockley-Read-Hall via defects, surface recombination) before collection → lower current and voltage	High-purity material, surface passivation, back-surface field, thin emitter
Resistive (electrical) losses	Series resistance (emitter sheet, contacts, fingers) and shunt (leakage) paths reduce the fill factor	Optimised grid, good contacts, avoid shunts
Collection and voltage losses	Not all carriers collected; open-circuit voltage less than E_g/q	Good junction design
Temperature losses	Efficiency falls ~0.4 %/°C for Si	Cooling, low-temperature-coefficient materials

Optical losses with schematic



Optical losses in a solar cell (drawn for this pack)

Optical losses are the part of the incident light that does not generate electron-hole pairs in the active layer. Three mechanisms:

1. **Reflection from the front surface**: bare silicon reflects 30-40 % of sunlight because of its high refractive index (n ≈ 3.5-4). Reduced by an **anti-reflection coating** (SiN_X, thickness λ/4n, n ≈ 2, gives destructive interference of reflected waves) and **surface texturing** (pyramids: reflected light hits the surface again).
2. **Shading by front metal contacts**: the front grid fingers and busbars cover 5-10 % of the surface and block light. Reduced by narrow, high fingers, buried contacts, or moving all contacts to the rear (IBC cells).
3. **Transmission / incomplete absorption**: silicon is an indirect-gap material that absorbs weakly near the band edge, so long-wavelength (infrared) photons may pass through a thin cell and leave through the back. Reduced by **light trapping**: texture + back reflector lengthen the optical path (up to $4n^2 \approx 50$ times).
4. Also: parasitic absorption in the ARC (anti-reflection coating), TCO (transparent conducting oxide) or rear metal (absorbed without creating useful carriers).

```
Optical efficiency: J_sc ∝ (1 - R)(1 - shading fraction) × absorbed fraction
```

> **Memory trick:** Losses = **"Optical, Spectral, Recombination, Resistive"**. Optical = **"Reflect, Shade, Transmit"**.

Section B: Supercapacitors, hydrogen storage, thin films, XRD and electron microscopy

B10 SEM: backscattered vs secondary electrons; how SEM gives high resolution and depth of field

Must study · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2022 Q8(a)** [6] Differentiate between backscattered electrons and secondary electrons in terms of their formation and role during image generation in SEM (scanning electron microscope).
- **2022 Q8(b)** [12] How does Scanning Electron Microscopy (SEM) enable the examination of microscopic structures with high resolution and depth of field?

★ REMEMBER IT

- **Secondary electrons**: low energy (< 50 eV), knocked out of the top few nm → **topography** (surface shape). **Backscattered electrons**: beam electrons bounced back, high energy → **composition contrast** (heavy atoms look bright).
- SEM depth of field is large because the beam is very narrow (tiny convergence angle), so a wide height range stays in focus.

Before you start: seeing with electrons

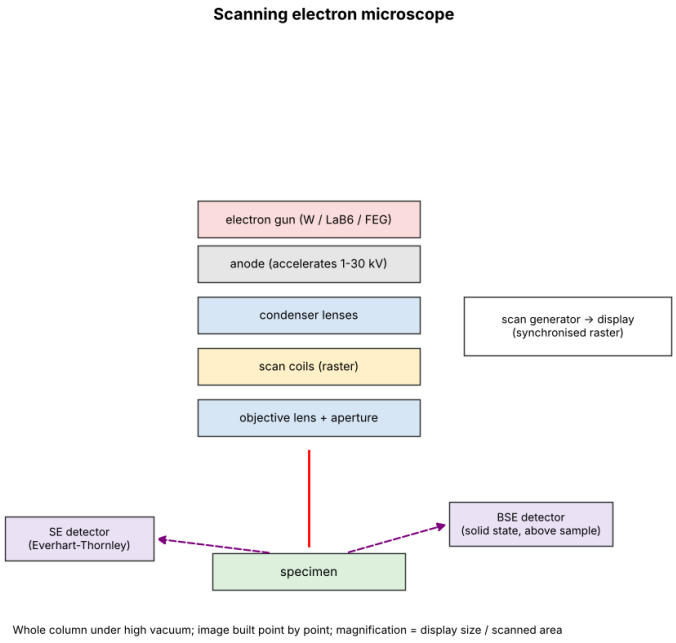
Light microscopes cannot resolve details smaller than ~200 nm (half the wavelength of light). Electrons accelerated by 1-30 kV have wavelengths far below 1 nm, so an electron microscope can see much finer detail.

In a **scanning electron microscope** (SEM), a fine electron beam is scanned across the surface like a raster on a TV screen. At each point, signals coming out of the sample are counted and shown as the brightness of that pixel. The two main signals are **secondary electrons** (SE) and **backscattered electrons** (BSE).

Backscattered vs secondary electrons

	Secondary electrons (SE)	Backscattered electrons (BSE)
Origin	Specimen electrons knocked out by **inelastic** collisions with the beam	Beam electrons scattered back by **elastic** collisions with nuclei (angle > 90°)
Energy	Low: < 50 eV (mostly 3-5 eV)	High: > 50 eV, up to the beam energy
Escape depth	Very shallow: ~1-10 nm	Deeper: ~0.1-1 µm (part of interaction volume)
Resolution	Highest (~1-5 nm)	Lower
Contrast	**Topographic** (surface shape, edges bright - edge effect)	**Compositional (atomic number, Z) contrast**: heavier elements appear brighter; also crystal orientation (channelling, EBSD (electron backscatter diffraction))
Detector	Everhart-Thornley detector (positively biased grid + scintillator + photomultiplier), in-lens detector	Solid-state (semiconductor) annular detector above the sample; or scintillator
Use	Surface morphology, fractography	Phase distribution, composition mapping, polished sections

Scanning electron microscope: construction and working



Schematic of a scanning electron microscope (drawn for this pack)

1. **Electron gun**: tungsten filament, LaB_6 or field-emission gun (FEG) emits electrons, accelerated by 1-30 kV through the anode.
2. **Condenser lenses** (electromagnetic): demagnify the beam to a small spot; apertures limit the beam.
3. **Scanning coils**: deflect the beam in a raster pattern over the specimen, synchronised with the display.
4. **Objective lens**: focuses the beam to a fine probe (~1-10 nm) on the sample surface; stigmators correct astigmatism.
5. **Specimen chamber and stage**: under high vacuum (~10^{-4}-10^{-6} torr); stage moves x, y, z, tilt, rotate. Non-conducting samples are coated with gold or carbon to stop charging.

6. **Beam-specimen interaction**: in a pear-shaped **interaction volume** the beam produces SE, BSE (backscattered electrons), characteristic X-rays (for EDS (energy-dispersive X-ray spectroscopy)/EDX chemical analysis), Auger electrons and cathodoluminescence.
7. **Detectors and display**: the detector signal is amplified and modulates the brightness of each pixel; magnification = display width / scanned width (10× to 500,000×). Large depth of field gives a 3-D look.

Side note: EDS = energy-dispersive X-ray spectroscopy: X-ray energies identify the elements present at the beam spot.

Memory trick: SE = Surface shape (low energy, shallow); BSE = Big atoms bright (Z contrast). SEM column: **"Gun, Condenser, Scan coils, Objective, Sample, Detector"**.

B6 Chemical vapour deposition (CVD): process and role in thin films, powders, single crystals; PVD vs CVD

Must study | Very likely · 100% | Easy

PAST-PAPER QUESTIONS

- **2022 Q6(c)** [10] Discuss the process of Chemical Vapor Deposition (CVD) and its role in the fabrication of thin films, powders, and single crystals.
- **2022 Q6(d)** [6] Compare and contrast between PVD (physical vapour deposition) and CVD (chemical vapour deposition) process.

★ REMEMBER IT

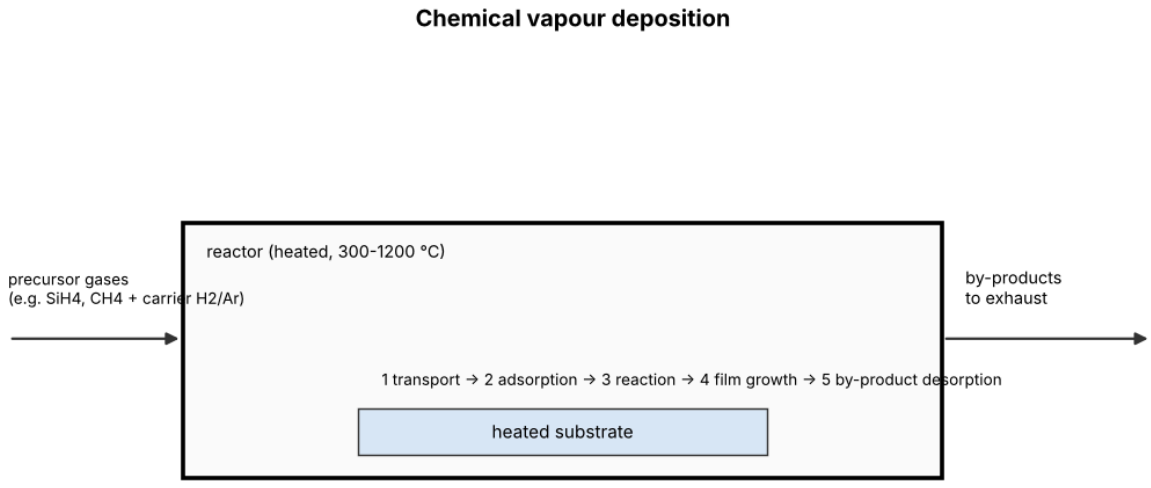
- PVD = **physical**: material is evaporated or sputtered and condenses (line of sight, lower T). CVD = **chemical**: precursor gases react on a hot substrate and deposit a solid (conformal coating, higher T, by-product gases). CVD steps: transport → adsorption → surface reaction → film growth → by-products desorb and leave.

Before you start: making films with chemistry

In **chemical vapour deposition** (CVD) the film is made by a chemical reaction: gas molecules (precursors) flow over a hot substrate, react or decompose on its surface, and leave a solid film; the by-product gases are pumped away. Example: SiH_4 (silane) → Si (film) + $2H_2$.

In **physical vapour deposition** (PVD) atoms are physically moved from a solid source (by evaporation or sputtering) onto the substrate, with no chemical reaction needed.

Chemical vapour deposition process



CVD reactor and steps (drawn for this pack)

1. **Transport of precursors**: reactant gases (with carrier gas, e.g. H_2, N_2, Ar) are fed into the reaction chamber and flow toward the substrate.
2. **Diffusion** through the boundary layer to the substrate surface.
3. **Adsorption** of reactants on the heated substrate surface.

4. **Surface reaction**: decomposition or reaction (heterogeneous), surface diffusion to growth sites, nucleation and film growth.
5. **Desorption** of by-products and their diffusion/transport out of the chamber to the exhaust (scrubbed if toxic).

Examples: $SiH_4 \rightarrow Si + 2H_2$; $SiH_4 + 2N_2O \rightarrow SiO_2 + 2N_2 + 2H_2$; $3SiH_4 + 4NH_3 \rightarrow Si_3N_4 + 12H_2$; $WF_6 + 3H_2 \rightarrow W + 6HF$

Energy sources / variants: thermal (APCVD - atmospheric pressure, LPCVD (low-pressure CVD) - low pressure), **PECVD (plasma-enhanced chemical vapour deposition)** (plasma-enhanced, low temperature ~200-400 °C: a-Si:H and SiN_x for solar cells), MOCVD (metal-organic: III-V multijunction cells, LEDs), ALD (atomic layer deposition, one layer at a time), hot-wire CVD.

Rate: at low temperature it is **surface-reaction-limited** (rate grows exponentially with T, Arrhenius); at high temperature **mass-transport (diffusion)-limited**.

PVD vs CVD

Aspect	PVD	CVD
Principle	Physical: evaporation or sputtering of a solid source, condensation	Chemical reaction / decomposition of gaseous precursors on the substrate
Source	Solid target / evaporant	Gases or volatile liquids
Temperature	Low-moderate (room T-500 °C)	Usually high (300-1200 °C), lower for PECVD
Pressure	High vacuum	Atmospheric to low pressure
Coverage	Line-of-sight; poor step coverage on complex shapes	Conformal, good step coverage, coats complex shapes and inside holes
Film	Dense, high purity, metals/alloys/compounds; good adhesion	High purity, crystalline/epitaxial possible, controllable stoichiometry
By-products / safety	None or few; clean	Toxic, corrosive or flammable gases (SiH_4, HF) need treatment
Deposition rate	Moderate	Can be high
Examples	Thermal/e-beam evaporation, sputtering, PLD	APCVD (atmospheric-pressure CVD), LPCVD, PECVD, MOCVD (metal-organic CVD), ALD

Memory trick: CVD steps = **"Transport, Diffuse, Adsorb, React, Desorb"**. PVD = **physical, line-of-sight**; CVD = **chemical, conformal**.

B5 Sputtering: definition and process Must study Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2022 Q6(b)** [9] Define sputtering. Describe the sputtering process with a suitable figure.

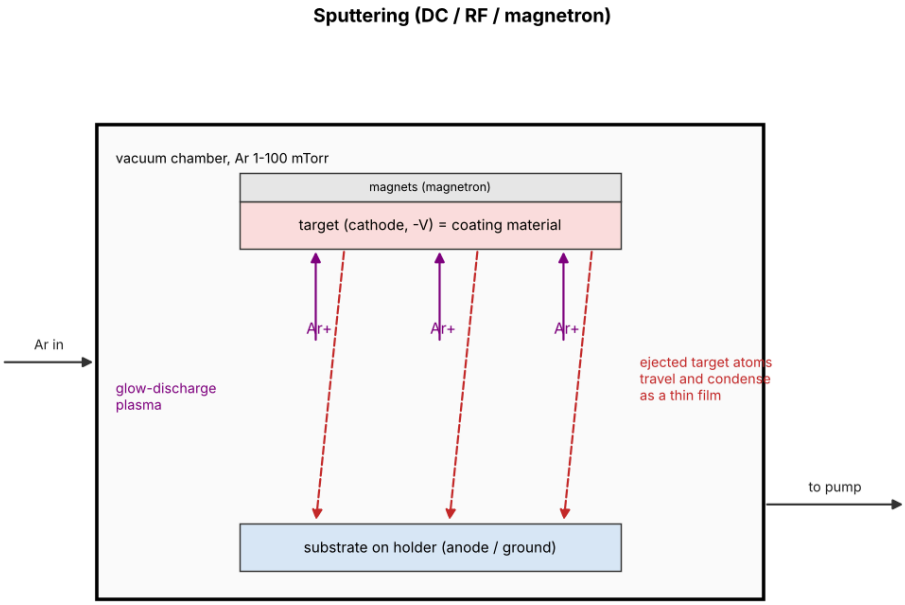
★ REMEMBER IT

- Sputtering = **atomic sandblasting**: Ar^+ ions from a plasma hit a target and knock atoms off; they fly to the substrate and form a thin film. Magnetron sputtering uses magnets to trap electrons near the target for a denser plasma.

Before you start: making thin films by atomic billiards

Sputtering is a physical vapour deposition (PVD) method. Fast heavy ions (usually argon, Ar^+) are fired at a solid **target** of the material you want; they knock atoms off it like billiard balls, and those atoms fly across and coat the **substrate**. No melting needed - works for metals, alloys, oxides, TCO (transparent conducting oxide) for solar cells.

Sputtering process



Sputtering system (magnetron, drawn for this pack)

1. **Vacuum**: the chamber is pumped to a high vacuum (~10^{-6} torr) to remove contaminants, then back-filled with inert **argon** to ~1-100 mtorr.
2. **Plasma (glow discharge)**: a high negative voltage (several hundred V) is applied to the target (cathode); the substrate holder/chamber is the anode. Free electrons accelerate, ionise Ar atoms: $Ar + e^- \rightarrow Ar^+ + 2e^-$, sustaining a plasma.
3. **Ion bombardment**: Ar^+ ions are accelerated across the cathode sheath and strike the target with high kinetic energy.
4. **Ejection**: momentum transfer through a collision cascade ejects (sputters) target atoms; also secondary electrons are emitted that keep the plasma going. **Sputter yield** = atoms ejected per incident ion (~0.1-3).

5. **Transport and deposition**: sputtered atoms travel through the gas and condense on the substrate, nucleate and grow into a dense, adherent thin film.

Types

Type	Feature
DC sputtering	DC voltage; only for conducting targets (insulators charge up)
RF sputtering	13.56 MHz alternating field; for insulating targets (oxides, nitrides)
Magnetron sputtering	Magnets behind target trap electrons near its surface → denser plasma, higher rate, lower pressure, less substrate heating (most used industrially)
Reactive sputtering	Reactive gas (O_2, N_2) added to form compounds, e.g. TiN, ZnO, ITO (indium tin oxide)
Ion-beam sputtering	Separate ion gun; precise control

Advantages: any material including high-melting metals and alloys (composition preserved), good adhesion and uniformity over large areas, dense films, room-temperature process. **Disadvantages:** lower deposition rate than evaporation, expensive targets, substrate heating and damage by energetic particles, gas incorporation.

> **Memory trick:** Sputtering steps: **"Vacuum, Argon, Plasma, Bombard, Eject, Deposit"**.

B8 Geometry and components of an X-ray diffractometer Must study

Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2022 Q7(b)** [9] Draw a geometric arrangement of an X-ray diffractometer with the functions of each component.

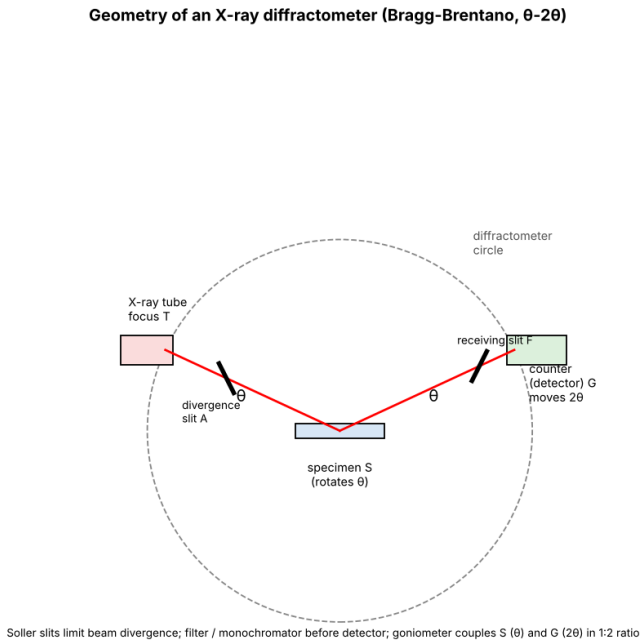
★ REMEMBER IT

- Diffractometer path: **X-ray tube → divergence (Soller) slit → sample on goniometer (θ) → receiving slit → filter/monochromator → detector (2θ)**. Detector moves twice as fast as the sample.

Before you start: the XRD (X-ray diffraction) machine

An X-ray **diffractometer** measures the intensity of X-rays diffracted by a powder or polycrystalline sample as a function of angle 2θ. Each peak position gives a d-spacing (Bragg's law), and the set of peaks identifies the crystal phase (fingerprint, compared with the ICDD (International Centre for Diffraction Data)/JCPDS database). Peak widths give crystallite size (Scherrer); intensities give structure and texture.

Construction



Bragg-Brentano diffractometer geometry (drawn for this pack)

Part	Function
X-ray tube (source S, line focus)	Produces X-rays from a Cu (or Mo, Co) target; high-voltage generator, water cooling
Soller slits and divergence slit	Limit axial and vertical divergence of the incident beam
Sample (specimen) C	Flat powder sample on the centre of the goniometer, rotates by θ
Goniometer	Precision circle carrying sample and detector; detector arm moves at 2θ (twice the sample speed)
Receiving slit F, anti-scatter slit, filter/monochromator	Define the diffracted beam, remove Kβ and background
Detector	Proportional, scintillation or solid-state (strip/area) counter; counts photons
Electronics and computer	Count rate vs 2θ: the diffractogram

Working (Bragg-Brentano θ-2θ geometry)

1. X-rays diverge from the line focus S of the tube, pass through the slits and fall on the flat sample C.
2. Sample rotates by θ while the detector rotates by 2θ on the diffractometer circle about the same axis; source, sample and receiving slit lie on a **focusing circle**, so diffracted rays from the whole irradiated area converge on the receiving slit (para-focusing).
3. Only crystallites whose planes are parallel to the sample surface and satisfy **nλ = 2d sinθ** diffract into the detector at that angle.
4. The detector records intensity as the angle is scanned (e.g. 2θ = 10°-90°); a peak appears at each Bragg angle.
5. Pattern analysis: peak positions → d-spacings → phase identification and lattice parameters; intensities → quantitative phase analysis and structure (Rietveld); widths → crystallite size t = 0.9λ/(B cosθ) and strain.

Advantages over film (Debye-Scherrer) methods: fast, accurate intensity measurement, direct digital data, high resolution.

> **Memory trick:** Diffractometer: **"Tube, Slits, Sample (θ), Detector (2θ), on a circle"**.

B7 Obtaining monochromatic Kα radiation from the X-ray tube spectrum

Must study | Very likely · 100% | Easy

PAST-PAPER QUESTIONS

- **2022 Q7(a)** [8] Explain how you will obtain monochromatic Kα radiation from the continuous X-rays and other characteristic X-rays.

★ REMEMBER IT

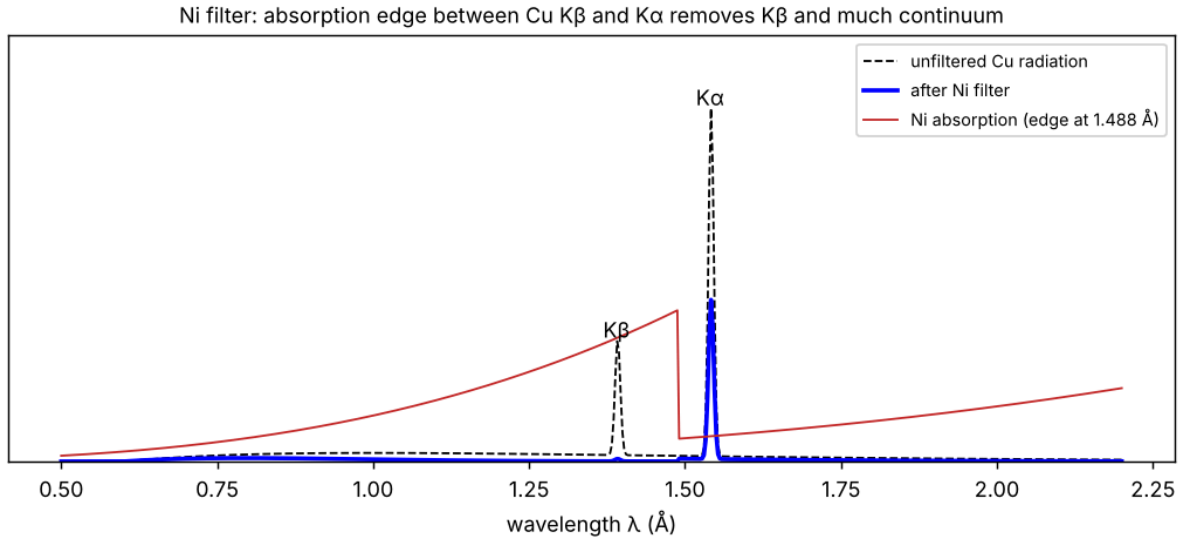
- Monochromatic Kα: use a **filter** whose absorption edge lies between Kα and Kβ of the target (Ni filter for Cu radiation) to kill Kβ and most of the white radiation, or a crystal **monochromator** set at the Kα Bragg angle.

Before you start: why one X-ray wavelength

An X-ray tube gives two kinds of X-rays: a smooth **continuous** spectrum (bremsstrahlung: braking radiation from decelerated electrons) and sharp **characteristic** lines (Kα, Kβ) when electrons knocked out of the inner K shell are replaced by electrons from outer shells (L → K gives Kα, M → K gives Kβ).

Diffraction uses Bragg's law nλ = 2d sinθ; to get one d for each peak we need **one known λ**, so we keep Kα and remove Kβ and the continuum. For a Cu target Kα = 1.5418 Å, Kβ = 1.3922 Å.

Obtaining monochromatic Kα radiation



X-ray spectrum and the effect of a Ni filter on Cu radiation (drawn for this pack)

1. **Choose the tube voltage** above the K-excitation voltage of the target (e.g. 30-45 kV for Cu, K excitation 8.98 kV) so strong K lines appear on top of the continuum; Kα is ~5-10 times stronger than Kβ.
2. **β-filter (absorption-edge filter)**: a thin foil of an element whose atomic number is one (or two) less than the target is placed in the beam. Its K absorption edge lies **between** the Kα and Kβ wavelengths of the target, so it absorbs Kβ (and short-wavelength continuum) strongly while passing most Kα. For Cu radiation: **Ni filter** (K edge 1.488 Å, between 1.392 and 1.542 Å), ~15-20 μm thick reduces Kβ/Kα from ~1/7 to ~1/500 while transmitting ~50 % of Kα. (Mo → Zr filter; Co → Fe; Fe → Mn; Cr → V.)

3. **Crystal monochromator**: a single crystal (graphite, Si, Ge, LiF) is set at the Bragg angle for Kα; only Kα is diffracted into the detector or onto the sample, removing Kβ, continuum and sample fluorescence. Placed in the incident or diffracted beam. (Better purity; separating $K\alpha_1$ from $K\alpha_2$ needs a high-resolution monochromator such as Ge(111) Johansson.)
4. Other: pulse-height discrimination in energy-sensitive detectors, balanced (Ross) filters, multilayer mirrors.

Side note: Why the edge works: an atom absorbs X-rays strongly just on the short-wavelength side of its K edge (photon energy enough to eject its K electron). Kβ is shorter than the Ni edge so it is absorbed; Kα is longer so it passes.

Memory trick: "Filter Z − 1": Cu (29) target → Ni (28) filter. Edge sits between Kβ and Kα.

B3 Pseudocapacitor vs double-layer capacitor

Must study · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2022 Q5(c)** [7] "Pseudocapacitor is a kind of supercapacitor which is different from double layer capacitor" - explain why.

★ REMEMBER IT

- EDLC (electric double-layer capacitor) = **physical** (ions line up, no reaction, non-Faradaic). Pseudocapacitor = **chemical** (fast surface redox, Faradaic). Battery = slow bulk chemistry. Speed: EDLC > pseudo > battery; energy: battery > pseudo > EDLC.

Before you start: capacitors that also use chemistry

An EDLC stores charge only physically on the surface. A **pseudocapacitor** also uses fast, reversible **redox** (reduction-oxidation, electron-transfer) reactions at or near the surface of the electrode. Electrons really cross the interface (like a battery) but the reaction is so fast and shallow that the device behaves electrically like a capacitor (charge proportional to voltage).

Pseudocapacitors: charge storage mechanism

A **pseudocapacitor** stores charge by **Faradaic** processes: fast, reversible electron-transfer reactions between the electrode and electrolyte ions, occurring at the surface or in a thin near-surface region. Unlike batteries, there is no slow phase change in the bulk, so the stored charge varies nearly linearly with potential (dQ/dV ≈ constant = "pseudo" capacitance). Three mechanisms (Conway):

Mechanism	What happens	Example
Underpotential deposition (adsorption)	Metal ions form an adsorbed monolayer on a different metal surface above their normal deposition potential	H on Pt, Pb on Au
Redox pseudocapacitance	Ions adsorb at the surface with charge transfer; surface atoms change oxidation state	$RuO_2 + xH^+ + xe^- \leftrightarrow RuO_{2-x}(OH)_x$; $MnO_2 + C^+ + e^- \leftrightarrow MnOOC$ (C = H^+, Na^+, K^+)
Intercalation pseudocapacitance	Ions insert into tunnels/layers of the host without phase change	Nb_2O_5, MXenes (Ti_3C_2) with Li^+

Materials

- **Transition metal oxides**: RuO_2 (best, ~700-1000 F/g, but expensive), MnO_2 (cheap, eco-friendly), NiO, Co_3O_4, V_2O_5, Fe_3O_4.
- **Conducting polymers**: polyaniline (PANI), polypyrrole (PPy), PEDOT (poly(3,4-ethylenedioxythiophene)) - doped/undoped by ions during charging (swelling limits cycle life).
- Heteroatom-doped (N, O) carbons with surface functional groups; MXenes; metal nitrides.

Comparison

	EDLC	Pseudocapacitor	Battery
Storage	Electrostatic (non-Faradaic)	Fast surface Faradaic redox	Bulk Faradaic, phase change
Capacitance/energy	100-200 F/g	10-100 times higher than EDLC per area	Highest energy
Power and life	Highest power, 10^6 cycles	High power, 10^3-10^5 cycles	Lower power, 10^3 cycles
CV curve	Rectangular	Nearly rectangular with broad peaks	Sharp redox peaks

Hybrid capacitors combine an EDLC carbon electrode with a pseudocapacitive or battery-type electrode (e.g. Li-ion capacitor) to get more energy with high power.

> **Memory trick: EDLC = physics (ions park); pseudo = fast surface chemistry (redox)**. Materials: **"RuO_2, MnO_2, conducting polymers"**.

B2 Numerical: capacitors 15 F and 8 F in series and in parallel across 50 V

Good to have | Very likely · 100% | Easy

PAST-PAPER QUESTIONS

- **2022 Q5(b)** [6] If two capacitors C_1 = 15 F and C_2 = 8 F are connected in series and in parallel to a 50 V battery, determine the voltage across each capacitor for each type of connection.

CALCULATOR **Two capacitors in series and in parallel**

Change any number; every step is recalculated and shown with the numbers put in.

1. Series C_{eq}: $C_1C_2/(C_1+C_2)$ = **15** × **8** / (**15** + **8**) = **5.217** F
2. Charge (same on both): $Q = C_{eq}V$ = **5.217** × **50** = **260.87** C
3. V across C_1: **260.9** / **15** = **17.39** V
4. V across C_2: **260.9** / **8** = **32.61** V
5. Parallel C_{eq}: **15** + **8** = **23.00** F
6. Parallel charge on C_1: **15** × **50** = **750.0** C
7. Parallel charge on C_2: **8** × **50** = **400.0** C
8. Energy, series: ½CV^2 = 0.5 × **5.217** × **50** × **50** = **6522** J
9. Energy, parallel: 0.5 × **23** × **50** × **50** = **28750** J

Before you start: capacitors in series and parallel

Capacitors combine the **opposite way** to resistors. In **series**, the same charge Q sits on each capacitor and the voltages add: $1/C_{eq} = 1/C_1 + 1/C_2$. In **parallel**, each capacitor gets the full voltage and the charges add: $C_{eq} = C_1 + C_2$.

Always: Q = C V. In series, the **smaller** capacitor takes the **larger** voltage.

Given: C_1 = 15 F, C_2 = 8 F, V = 50 V.

(i) Connected in series

Step (write this)	Why / what it means
$C_{eq} = C_1C_2/(C_1 + C_2) = (15 \times 8)/(15 + 8) = 120/23 =$ **5.217 F**	Product over sum (two capacitors in series).
$Q = C_{eq}V = 5.217 \times 50 =$ **260.87 C**	The same charge flows onto both capacitors in series.
$V_1 = Q/C_1 = 260.87/15 =$ **17.39 V**	Voltage across the 15 F capacitor.
$V_2 = Q/C_2 = 260.87/8 =$ **32.61 V**	Voltage across the 8 F capacitor (smaller C gets more voltage).
Check: 17.39 + 32.61 = 50 V ✓	Series voltages must add up to the supply.
Shortcut: $V_1 = V\,C_2/(C_1 + C_2) = 50 \times 8/23 = 17.39$ V	Voltage divider for capacitors uses the **other** capacitor on top.

(ii) Connected in parallel

Step (write this)	Why / what it means
C_{eq} = 15 + 8 = **23 F**	Parallel capacitances simply add.
V_1 = V_2 = **50 V**	Both are connected directly across the supply.
Q_1 = 15 × 50 = 750 C; Q_2 = 8 × 50 = 400 C; Q_{total} = 1150 C	Extra: charge on each.

Connection	C_{eq}	V across 15 F	V across 8 F
Series	5.217 F	17.39 V	32.61 V
Parallel	23 F	50 V	50 V

Side note: Energy stored (extra): series E = ½ × 5.217 × 50^2 = 6.52 kJ; parallel E = ½ × 23 × 50^2 = 28.75 kJ.

Memory trick: "Capacitors are opposite to resistors": series → product/sum, parallel → add.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2022 5(b) Given: C_1 = 15 F, C_2 = 8 F, V = 50 V

(i) Series: $C_{eq} = C_1C_2/(C_1 + C_2)$

= 120/23

= **5.217 F**

Charge: $Q = C_{eq}V$

= 5.217 × 50

= **260.9 C**

$V_1 = Q/C_1$ = 260.9/15 = **17.39 V**; V_2 = 260.9/8 = **32.61 V** (check 17.39 + 32.61 = 50 V)

(ii) Parallel: $C_{eq} = C_1 + C_2$

= 15 + 8

= **23 F**

$V_1 = V_2$ = **50 V**; Q_1 = 750 C, Q_2 = 400 C

B4 Hydrogen storage in carbonaceous materials and CNTs: adsorption mechanism, key parameters, applications

Must study · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

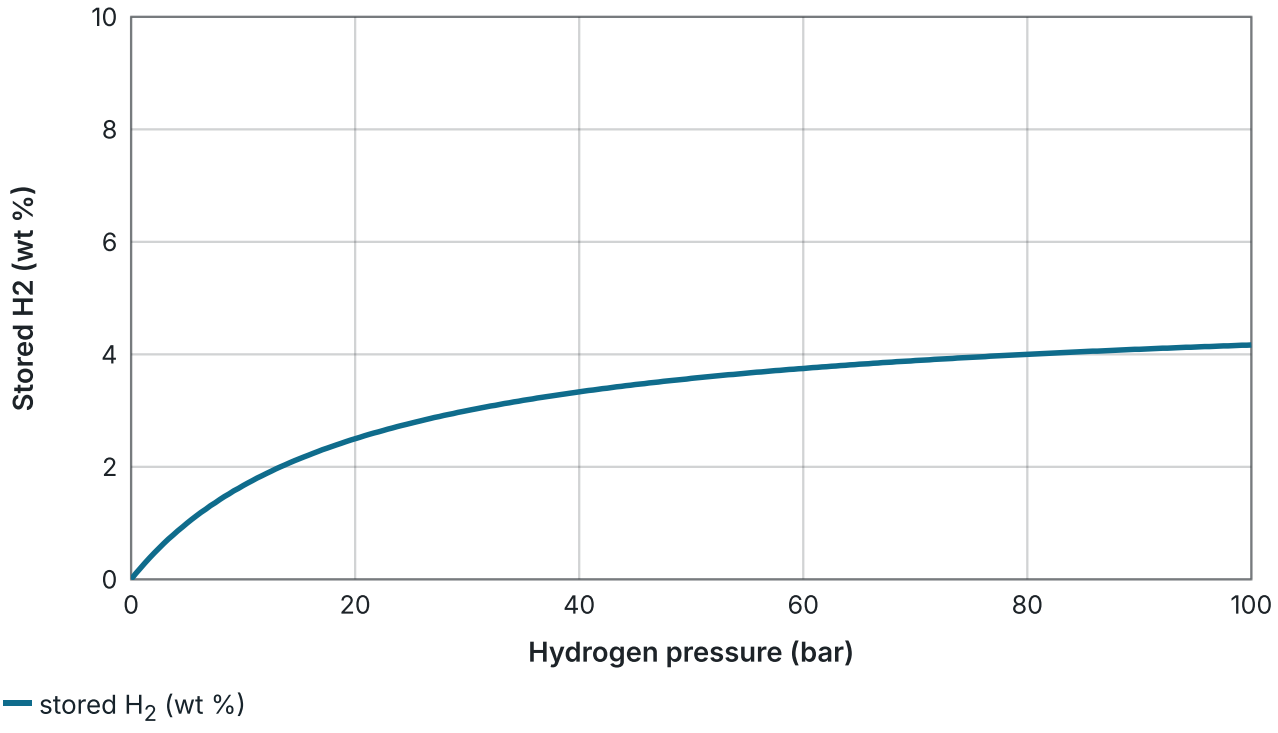
- **2022 Q5(d)** [10] Describe the hydrogen adsorption mechanism in carbonaceous materials. What are the key parameters for increasing H_2 storage capacity in CNTs?
- **2022 Q6(a)** [10] How do carbonaceous materials such as carbon nanotubes contribute to hydrogen storage and what are their potential applications in this field?

UNDERSTAND **Read this to understand; you do not need to write it**

Hydrogen sticks to carbon surfaces by weak van der Waals forces (**physisorption**): no bond breaks, so it is reversible but needs low temperature (77 K) or high pressure. More surface area, the right pore size (~0.7 nm), and doping or metal decoration (which strengthens binding) raise the capacity. CNT (carbon nanotube): H_2 sits outside, inside the tubes and in the gaps between bundled tubes.

INTERACTIVE **Langmuir adsorption isotherm: how much H_2 is stored vs pressure**

Move the sliders and watch the curve change.



Stronger binding (bigger K: cold, doped carbon) fills the surface at lower pressure. The DOE target is about 5.5 wt % at room temperature.

Before you start: hydrogen sticking to carbon

Hydrogen is a very light gas, so storing enough of it in a small tank is hard (needs 700 bar or -253 °C). Another way is to make H_2 molecules **stick (adsorb)** onto a solid with a huge surface area - like a sponge.

Physisorption = weak van der Waals attraction (~4-10 kJ/mol), works well only at low temperature (77 K, liquid nitrogen). **Chemisorption** = real chemical C-H bonds, strong, but needs heat to release the hydrogen. Target (US DOE): 5.5 wt % on a system basis.

Hydrogen storage in carbon materials

Carbon materials (activated carbon, carbon nanotubes, graphene, carbon nanofibres, templated and carbide-derived carbons, fullerenes) are light, cheap, porous and chemically stable, so they are candidates for adsorbing hydrogen:

- **Physisorption**: H_2 molecules held on the carbon surface by van der Waals forces; capacity proportional to specific surface area (Chahine rule: ~1 wt % per 500 m^2/g at 77 K). High-surface activated carbons (~3000 m^2/g) store ~5-7 wt % at 77 K, but < 1 wt % at room temperature because binding is too weak. Fast and fully reversible.
- **Chemisorption**: atomic hydrogen forms C-H bonds (e.g. hydrogenated graphene, graphane, up to ~7.7 wt % theoretically); release needs high temperature (slow, energy-costly).
- **Spillover**: metal nanoparticles (Pt, Pd) dissociate H_2 and the H atoms migrate onto the carbon support, increasing room-temperature storage.

Possible ways to increase H_2 storage capacity

1. **Increase specific surface area** and micropore volume (chemical activation with KOH, templated carbons, graphene aerogels).
2. **Optimise pore size** to ~0.6-0.7 nm (about two H_2 molecular diameters): overlapping wall potentials raise binding energy.
3. **Heteroatom doping** (B, N) to raise binding energy toward the ideal ~15-25 kJ/mol for room-temperature storage.
4. **Metal decoration** (Li, Ti, Ca, Pd, Pt) for spillover and Kubas-type binding.
5. **Lower temperature / higher pressure** operation (cryo-adsorption at 77 K).
6. Composites with metal hydrides or MOFs; open CNT ends, purify them and intercalate between graphene layers.

Carbon nanotubes (CNTs) for hydrogen storage

Structure: a CNT is a graphene sheet rolled into a seamless cylinder, ~1-2 nm diameter (single-walled, SWCNT (single-walled carbon nanotube)) or several concentric tubes (multi-walled, MWCNT (multi-walled carbon nanotube)). They are made by arc discharge, laser ablation or catalytic CVD (chemical vapour deposition).

Storage sites: hydrogen can adsorb (1) on the outer surface, (2) inside the tube (opened ends), (3) in interstitial channels between tubes in a bundle, (4) in grooves on the bundle surface; plus chemisorption on curved walls.

Feasibility: early reports (Dillon 1997; Chambers 1998 claimed 67 wt % in nanofibres) raised great hopes, but careful later experiments gave < 1 wt % at room temperature and ~2-6 wt % at 77 K -

similar to activated carbon of equal surface area; errors came from impurities and measurement. Advantages: light, stable, reversible, fast kinetics. Problems: weak binding at ambient temperature, high cost, difficulty in purification and opening, low volumetric density. **Conclusion:** pure CNTs are not yet practical for vehicles at room temperature; doped/metal-decorated CNTs and cryogenic operation remain research directions.

> **Memory trick:** Raise H_2 in carbon = **"Area, Pore size, Dope, Decorate, Cool"**. CNT sites = **"outside, inside, between, grooves"**.

B9 XRD numericals: 2θ of Co (111) with Cu Kα; structure factor of an FCC cell

Must study · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2022 Q7(c)** [6] A crystal of cobalt (Co) is analysed using CuKα (λ = 1.5418 Å) X-ray radiation. Calculate the angle of reflection 2θ of the (1 1 1) plane. The lattice parameter of FCC cobalt is 3.544 Å.
- **2022 Q7(d)** [12] Calculate the structure factor for an FCC unit cell and show six planes from where reflection may occur or not.

LEARN IT FROM ZERO

XRD (X-ray diffraction) numbers, from zero

Atoms in a crystal sit in flat parallel layers (planes) a distance d apart. X-rays bouncing off neighbouring layers travel an extra distance 2d sinθ. When that extra distance is a whole number of wavelengths the waves add up and a bright peak appears: **Bragg's law nλ = 2d sinθ**. The machine records 2θ.

For a cubic crystal with side a, the spacing of plane (hkl) is **$d = a / \sqrt{h^2 + k^2 + l^2}$**. Bigger indices → closer planes → larger angle.

But Bragg's law only says a peak *may* appear. The **structure factor F** adds the waves from every atom in the unit cell; if they cancel, F = 0 and the peak is missing. FCC: F = 4f when h, k, l are all odd or all even, else 0. BCC: F = 2f when h + k + l is even, else 0.

CALCULATOR **Bragg angle and FCC/BCC peak check for any (hkl)**

Change any number; every step is recalculated and shown with the numbers put in.

1. $h^2 + k^2 + l^2$: **1** × **1** + **1** × **1** + **1** × **1** = **3**
2. Plane spacing: $d = a/\sqrt{(h^2+k^2+l^2)}$ = **3.544** / √(**3**) = **2.0461** Å
3. sin θ: sinθ = nλ/2d = **1** × **1.542** / (2 × **2.046**) = **0.37676**
4. Bragg angle θ: $\sin^{-1}$(**0.3768**) = **22.13** °
5. Diffraction angle 2θ: 2 × **22.13** = **44.27** °
6. FCC peak?: **present, F = 4f**
7. BCC peak?: **absent, F = 0 (h+k+l odd)**

If sinθ comes out above 1 the plane cannot diffract with this wavelength.

Before you start: the two XRD formulas you need

Bragg's law: $n\lambda = 2d \sin\theta$. X-rays reflected from parallel planes reinforce when the extra path $2d \sin\theta$ equals a whole number of wavelengths.

Cubic d-spacing: $d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}$, where a = lattice parameter and (hkl) = Miller indices of the plane.

Structure factor $F_{hkl} = \Sigma f_n e^{2\pi i(hu_n + kv_n + lw_n)}$ adds up the waves scattered by every atom in the unit cell. If F = 0, the reflection is **absent** even though Bragg's law is satisfied. Useful facts: $e^{n\pi i} = +1$ if n is even, -1 if n is odd.

(c) Diffraction angle of Co (111), FCC, a = 3.544 Å, Cu Kα λ = 1.5418 Å

Step (write this)	Why / what it means
$d_{111} = a/\sqrt{(1^2 + 1^2 + 1^2)} = 3.544/\sqrt{3} = 3.544/1.7321 =$ **2.0461 Å**	Spacing between neighbouring (111) planes.
$\sin\theta = n\lambda/(2d) = 1 \times 1.5418/(2 \times 2.0461) = 1.5418/4.0922 = 0.37676$	Bragg's law with first order n = 1.
$\theta = \sin^{-1}(0.37676) =$ **22.13°**	The Bragg angle.
$2\theta =$ **44.27°**	Diffractometers record 2θ (angle between incident and diffracted beam); this matches the known strongest Co peak at ~44.2°.

(d) Structure factor of FCC and reflections present

FCC has 4 identical atoms per cell at (0,0,0), (½,½,0), (½,0,½), (0,½,½).

Step (write this)	Why / what it means
$F = f[e^{2\pi i(0)} + e^{2\pi i(h/2 + k/2)} + e^{2\pi i(h/2 + l/2)} + e^{2\pi i(k/2 + l/2)}]$	Sum over the 4 atoms: substitute each (u, v, w) into hu + kv + lw.
$F = f[1 + e^{\pi i(h + k)} + e^{\pi i(h + l)} + e^{\pi i(k + l)}]$	Simplify $2\pi i \times (1/2) = \pi i$.
h, k, l unmixed (all even or all odd): h + k, h + l, k + l all even ⇒ each term = +1 ⇒ **F = 4f**, $F^2 = 16f^2$	Sum of two odd or two even numbers is even, $e^{\text{even}\cdot\pi i} = 1$.
h, k, l mixed (e.g. 1,0,0): two of the sums are odd ⇒ F = f(1 + 1 - 1 - 1) = **0**	Waves from the face atoms cancel those from the corner atom: reflection absent.

Plane (hkl)	Indices	F	Present?
(100)	mixed	0	Absent
(110)	mixed	0	Absent
(111)	all odd	4f	**Present**
(200)	all even (0 is even)	4f	**Present**
(210)	mixed	0	Absent
(211)	mixed	0	Absent
(220)	all even	4f	**Present**
(311)	all odd	4f	**Present**
(222)	all even	4f	**Present**

So FCC metals (Co, Cu, Al, Ni) show the sequence (111), (200), (220), (311), (222)...; $h^2 + k^2 + l^2$ = 3, 4, 8, 11, 12. (Compare BCC: present when h + k + l is even: (110), (200), (211)...)

> **Memory trick: FCC = "unmixed"** (all odd or all even) → 4f; **BCC = "sum even"** → 2f. Zero counts as even.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2022 7(c) Given: Co, FCC, a = 3.544 Å, Cu Kα λ = 1.5418 Å, plane (111), n = 1

Plane spacing: $d_{111} = a/\sqrt{(1^2 + 1^2 + 1^2)}$

$= 3.544/1.7321$

$=$ **2.0461 Å**

Bragg's law: $\sin\theta = n\lambda/(2d)$

$= 1.5418/(2 \times 2.0461) = 0.37676$

$=$ **θ = 22.13°**

2θ = 44.27° (matches the strongest Co peak near 44.2°)

7(d) $F = f[1 + e^{\pi i(h+k)} + e^{\pi i(h+l)} + e^{\pi i(k+l)}]$: unmixed ⇒ F = 4f; mixed ⇒ F = 0. Present: (111), (200), (220); absent: (100), (110), (210).

B11 TEM: diffraction pattern and image together; preparing a 0.1 mm ceramic sample

Must study | Very likely · 100% | Medium

PAST-PAPER QUESTIONS

- **2022 Q8(c)** [8] How are diffraction pattern and image simultaneously present in TEM (transmission electron microscope)?
- **2022 Q8(d)** [9] You need to analyse a ceramic sample (0.1 mm thick) in TEM. Explain how you will prepare the sample for TEM investigation.

★ REMEMBER IT

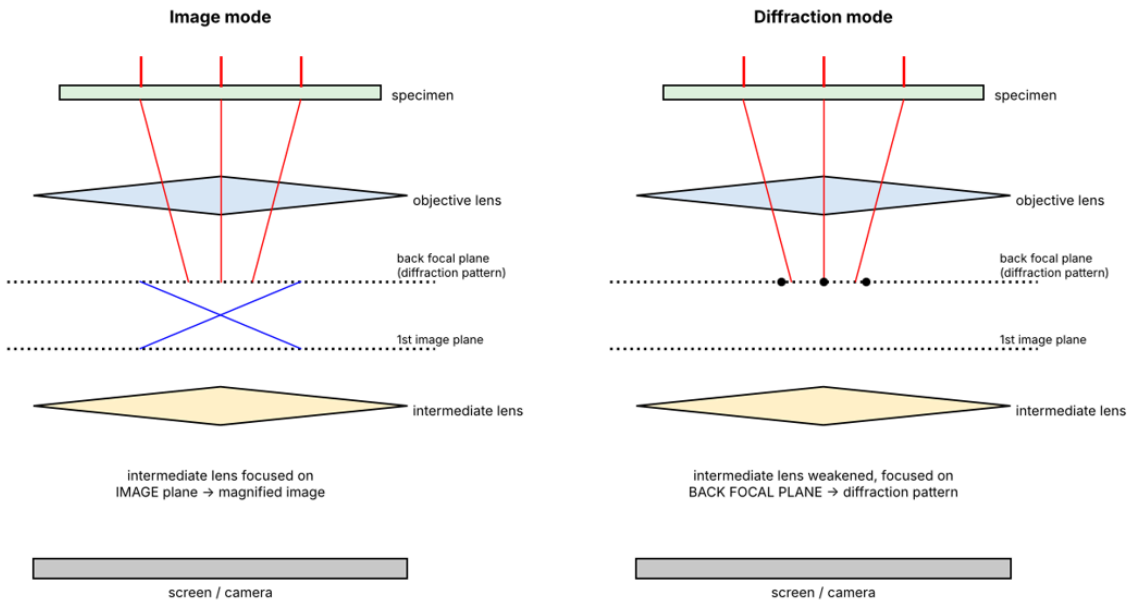
- TEM: below the specimen the objective lens forms both a **diffraction pattern** (in its back focal plane) and an **image** (in its image plane). Change the intermediate lens strength to look at one or the other.
- Ceramic 0.1 mm → electron transparent (< 100 nm): **cut → grind/polish to ~30 µm → dimple → ion mill (Ar^+) until a hole forms**; edges around the hole are thin enough. Or FIB (focused ion beam) lift-out.

Before you start: electrons through the sample

A **transmission electron microscope** (TEM) sends a high-energy (100-300 kV) electron beam **through** a very thin sample (< 100 nm). It can show atoms (resolution ~0.1 nm) and, like XRD (X-ray diffraction), produce diffraction patterns of tiny regions.

Every lens makes two important planes: the **back focal plane**, where all rays leaving the sample in the same direction meet (this is the **diffraction pattern**), and the **image plane**, where rays from the same point of the sample meet (this is the **image**). The TEM switches between modes by changing which one the next lens shows on the screen.

Imaging and diffraction modes in TEM



Imaging mode vs diffraction mode in a TEM (drawn for this pack)

	Imaging mode	Diffraction mode
Intermediate lens focused on	**Image plane** of the objective lens	**Back focal plane** of the objective lens
Aperture used	**Objective aperture** in the back focal plane (selects beams)	**Selected-area aperture** in the image plane (selects region)
Screen shows	Magnified image of the specimen	Diffraction pattern (spots for single crystal, rings for polycrystal, diffuse halo for amorphous): SAED (selected-area electron diffraction)
Information	Morphology, defects (dislocations, grain boundaries), particle size, lattice imaging (HRTEM)	Crystal structure, orientation, lattice spacings ($R\,d = \lambda\,L$, L = camera length)

Image contrast in imaging mode: Bright-field - objective aperture passes only the direct (transmitted) beam; diffracting or thick/heavy regions look dark. **Dark-field** - aperture passes one diffracted beam only; regions that diffract into that beam look bright. High-resolution TEM - many beams interfere (phase contrast) to image lattice fringes.

Sample preparation of a ceramic for TEM

Ceramics are hard, brittle and usually non-conducting, so they are thinned mechanically and then by ion milling until electron-transparent (< 100 nm):

Step (write this)	Why / what it means
1. Cut a thin slice (~0.5-1 mm) with a low-speed diamond saw	Diamond is needed for hard ceramics; slow cutting avoids cracks.
2. Core a **3 mm disc** with an ultrasonic or abrasive disc cutter	Standard TEM holder size.
3. Grind and polish both faces to ~100 µm with diamond lapping films	Removes saw damage; gives flat, parallel faces.
4. **Dimple grind** the centre to ~10-20 µm	Thins the centre while the thick rim keeps strength.
5. **Argon ion milling** (4-6 kV, low angle 4-8°) until a small hole forms; thin edges around it are electron-transparent	Ions sputter atoms gently away; ceramics cannot be electropolished as they are insulators.
6. **Carbon coat** the sample (thin layer)	Prevents charging of the insulating ceramic in the beam.
Alternatives: **FIB** (focused ion beam, Ga^+) lift-out of a site-specific lamella; crushing powder and dispersing on a holey-carbon grid; tripod polishing	FIB is fast and site-specific; crushing is simplest for powders.

> **Memory trick: "Back focal plane = Diffraction, Image plane = Image"**; the intermediate lens chooses. Ceramic prep: **"Saw, Disc, Grind, Dimple, Ion-mill, Coat"**.

B1 Electric double-layer capacitor (EDLC): why high capacitance, fundamental principle

Must study · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2022 Q5(a)** [12] Why does an Electric Double Layer Capacitor (EDLC) show high capacitance? Describe the fundamental principles of EDLC (electric double-layer capacitor) with necessary figure.

LEARN IT FROM ZERO

Supercapacitors, from zero

A normal capacitor stores charge on two plates separated by an insulator: C = εA/d. To store a lot you need a huge area A and a tiny gap d.

An **electric double-layer capacitor (EDLC)** gets both: its electrodes are activated carbon with an enormous internal surface (1000–3000 m^2 per gram), and the "gap" is just a layer of ions in the electrolyte sitting against the carbon surface, less than a nanometre thick. Charge is stored **physically** by ions lining up (no chemical reaction), so it charges in seconds and lasts a million cycles. Because d is so small and A so large, C reaches thousands of farads.

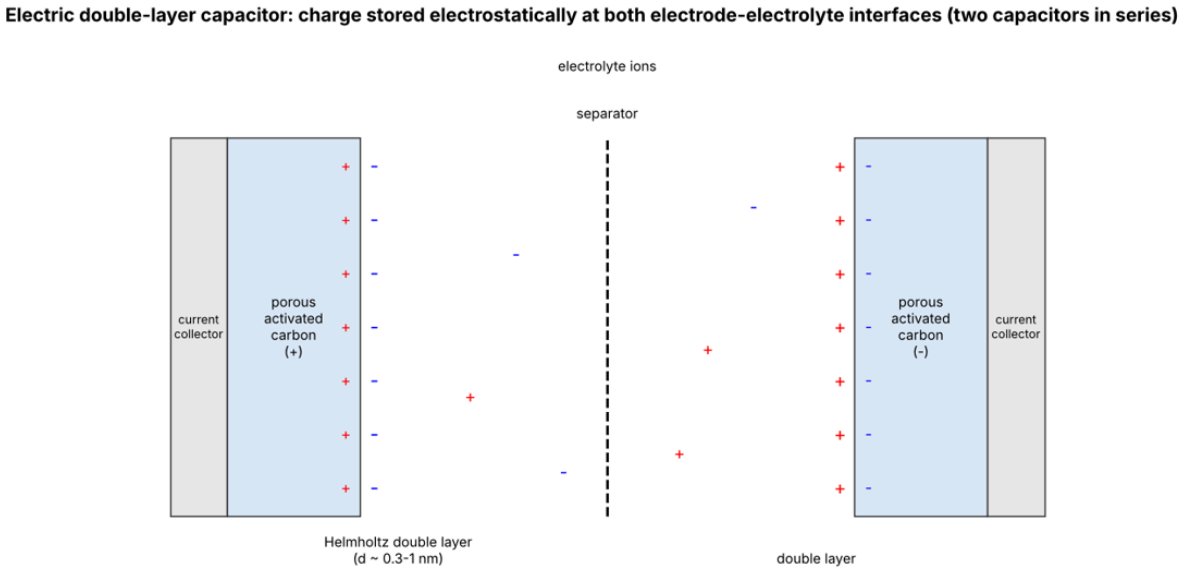
A **pseudocapacitor** also stores charge through fast **chemical** (redox) reactions at the surface of metal oxides (RuO_2, MnO_2) or conducting polymers: higher capacitance, but slower and shorter life.

Before you start: storing charge on a surface

A normal capacitor stores energy as charge on two metal plates separated by an insulator: C = εA/d. To store a lot, you need a huge area A and a tiny gap d.

A **supercapacitor** (ultracapacitor) does this with porous carbon (activated carbon has 1000-2000 m^2 of surface per gram!) soaked in an electrolyte. The "gap" is only the thickness of a layer of ions, less than 1 nm. Result: thousands of times more capacitance than a normal capacitor, and much faster charging than a battery.

Electric double-layer capacitor (EDLC)



Electric double-layer capacitor: charged state (drawn for this pack)

Construction: two porous high-surface-area **activated carbon** electrodes (also carbon nanotubes, graphene, carbide-derived carbon) coated on aluminium foil current collectors; a porous **separator** (paper or polymer) between them; the whole soaked in an **electrolyte** - aqueous (H_2SO_4, KOH; 1 V), organic ($TEABF_4$ in acetonitrile; 2.7 V) or ionic liquid (3.5-4 V).

Working (charge storage mechanism):

1. When a voltage is applied, one electrode becomes positive and the other negative.
2. Ions of opposite charge in the electrolyte move to each electrode surface: anions to the positive electrode, cations to the negative one.
3. At each electrode-electrolyte interface an **electric double layer** (Helmholtz layer) forms: the electronic charge in the carbon and a layer of ions in the electrolyte, separated by about 0.3-0.8 nm (solvent molecules).
4. Charge is stored **electrostatically** - no chemical (Faradaic) reaction, no electron transfer across the interface. So charge/discharge is very fast and reversible, > 10^5-10^6 cycles.
5. The cell is two double-layer capacitors in series: $1/C_{cell} = 1/C_+ + 1/C_-$; with equal electrodes $C_{cell} = C_{electrode}/2$.
6. On discharge, ions return to the bulk electrolyte and electrons flow through the external circuit.

$C = \varepsilon_r\varepsilon_0 A / d$ $\quad$ $E = \tfrac{1}{2}CV^2$ $\quad$ $P_{max} = V^2/(4\ ESR)$

> **Side note:** Large A (porous carbon) and tiny d (one ion layer) give 100-200 F per gram of carbon. Energy rises with V^2, so electrolytes with a wide voltage window (organic, ionic liquid) store more energy. ESR = equivalent series resistance.

Models of the double layer: Helmholtz (single compact layer of ions), Gouy-Chapman (diffuse layer), Stern (compact + diffuse).

Characteristics: high power density (5-10 kW/kg), modest energy density (5-10 Wh/kg, vs 100-250 for Li-ion), charging in seconds, very long life, wide temperature range, safe. Uses: regenerative braking, buses and trams, back-up memory, power smoothing for wind/solar, start-stop cars, camera flashes.

Memory trick: EDLC = **"big Area (porous carbon), tiny d (ion layer), no reaction"** → fast and long-life but low energy.